

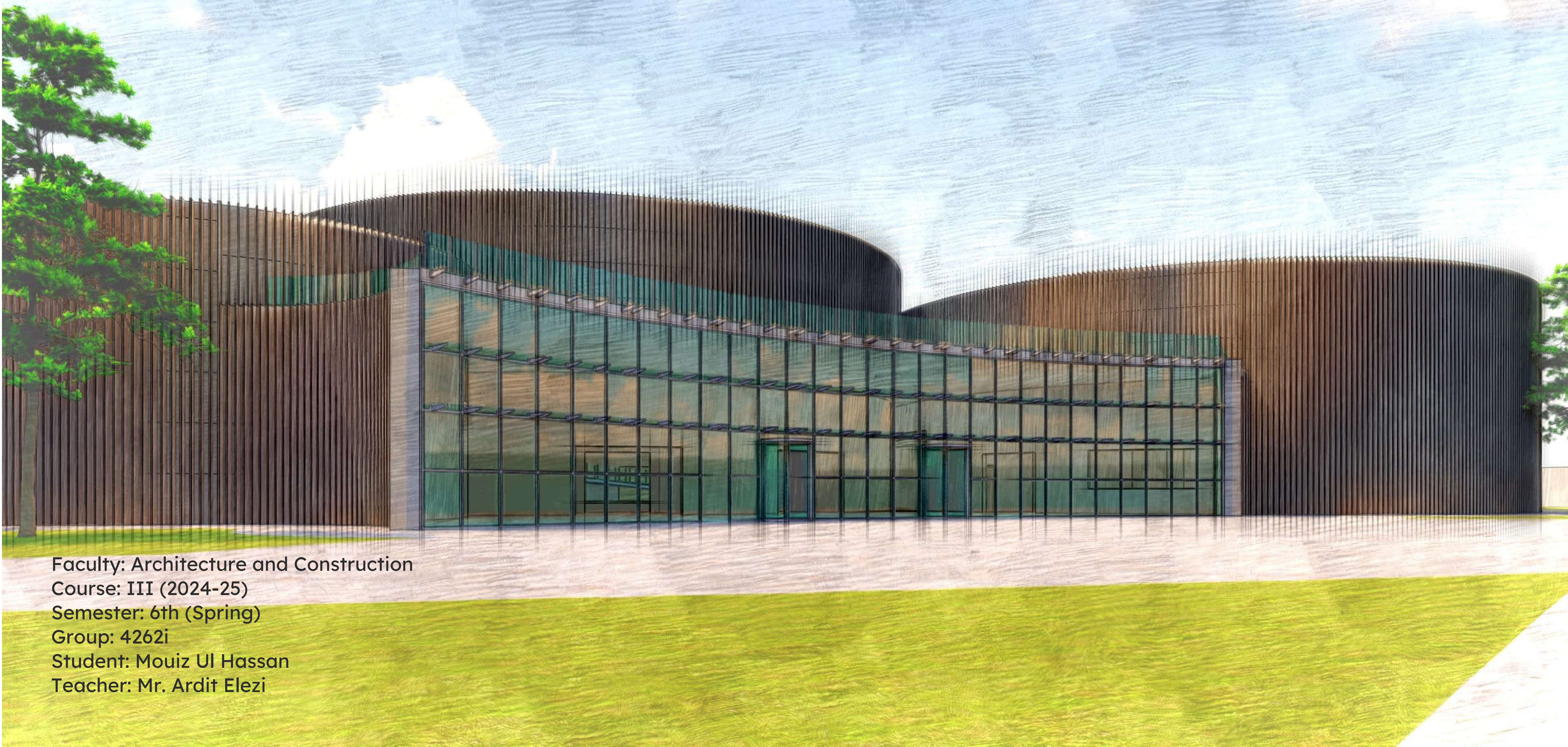
Course project
Architectural Design - 4

Design of
Caspian Oil & Gas Museum

Located in
Baku, Azerbaijan

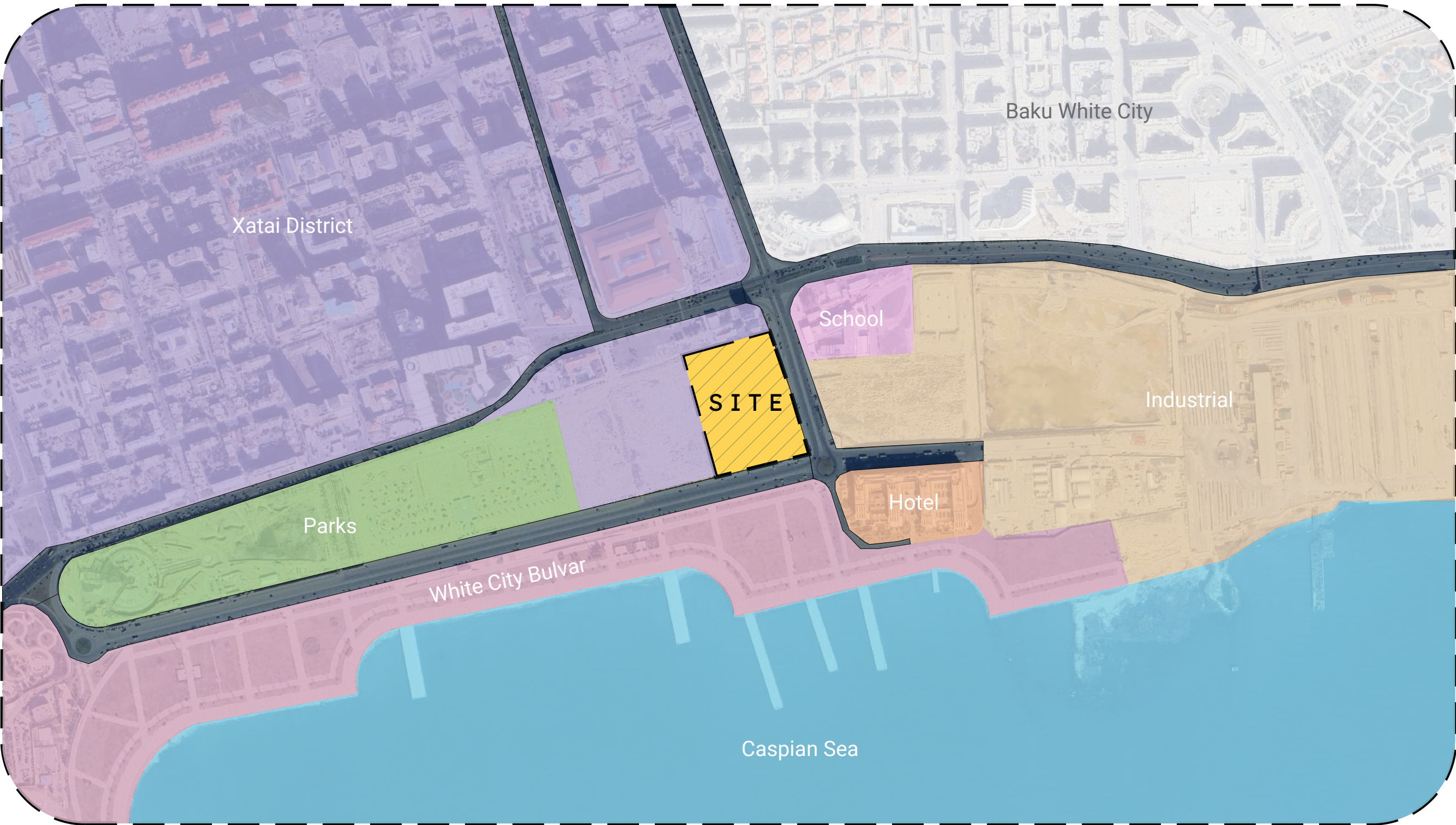


BMU
BAKİ MÜHƏNDİSLİK UNIVERSİTETİ



Faculty: Architecture and Construction
Course: III (2024-25)
Semester: 6th (Spring)
Group: 4262i
Student: Mouiz Ul Hassan
Teacher: Mr. Ardit Elezi

3. SITE ANALYSIS - Location, Accessibilty, Vegetation & Figure Ground



Location: 8 November Avenue near Hərbi Qənimətlər Parkı
Area: 23,600 m² | Perimeter: 660 m



Road Network

The metro System in Baku has only 2 lines which connect to major points of the city. The nearest metro station to the site is 20 January m/st which is located at distance of approx. 500 meters.

It can be used to travel to go 28 May from where a transfer to Içeri Seher or Jaffar Jabbarli line. It can also be used to go Memar Ajami from where one can transfer to Purple line connecting the international bus terminal.

Bus Network

Baku has a vast bus system consisting of petrol, gas and electric powered buses. They connect the city very well and most of the city has dedicated bus lanes which make it easier to commute in rush hours.

The site has 4 bus stops in close proximity and they serve 9 bus routes which connect to major points in Baku such as 20 Yanvar, Koroglu, Yeni Yasamal, Baku International Bus Terminal, Inshaatchlar, Qurd Qapisi, Nizami, Sederek, Nasimi, Bina Shopping Center, Lokbatan, Sahil, Içeri Seher & Highland Park to name a few.

The following bus routes are available:

10 20 21 32 46 72 88a 120 120E

Metro Network

The metro System in Baku has only 2 lines which connect to major points of the city. The nearest metro station to the site is Khatai metro station which is located at a distance of 1.5 km or 15 minutes by walk.

It can be used to travel to go 28 May from where a transfer to Içeri Seher or Jaffar Jabbarli line. It can also be used to go Memar Ajami from where one can transfer to Purple line connecting the international bus terminal.

Overview:

- The site currently lacks significant greenery, as it is part of a newly redeveloped area with minimal existing vegetation. The absence of trees and landscaped zones means that strategic planning will be necessary to integrate green spaces into the project. Given Baku's hot summers and strong winds, incorporating windbreaks, shaded courtyards, and rooftop gardens can enhance thermal comfort and visitor experience.
- Additionally, the proximity to White City Boulevard and the waterfront presents an opportunity to create green buffers and pedestrian-friendly landscapes, seamlessly connecting the museum to the surrounding urban fabric. Sustainable landscaping with native drought-resistant plants can further reduce water consumption while improving the aesthetic and ecological quality of the site.



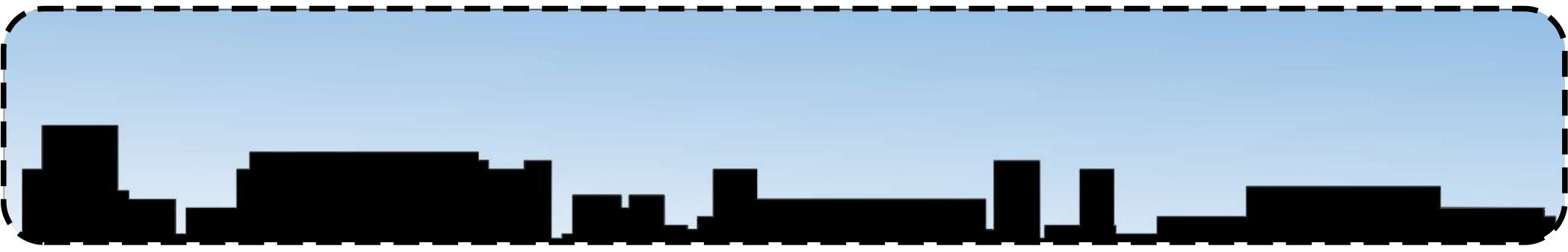
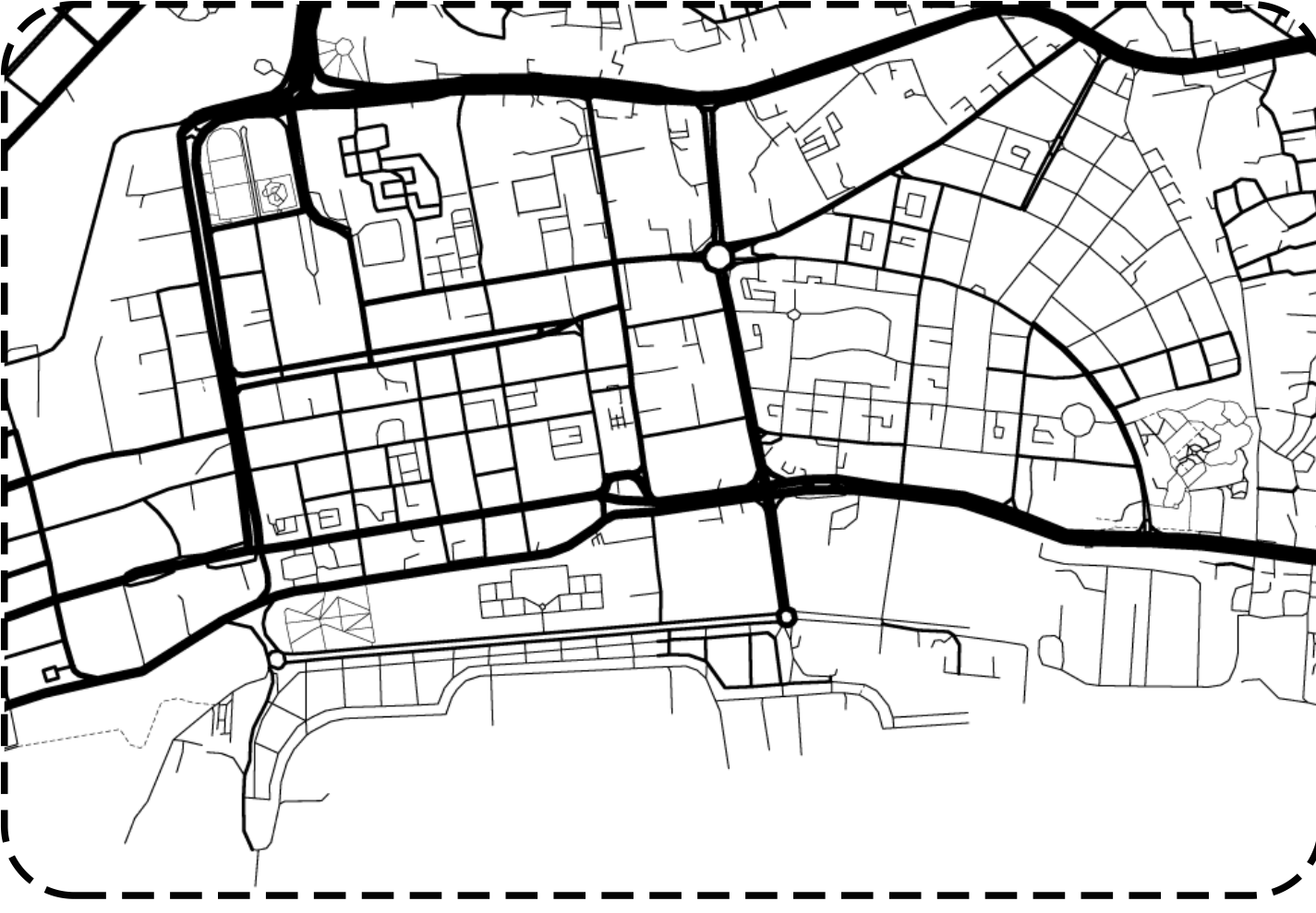
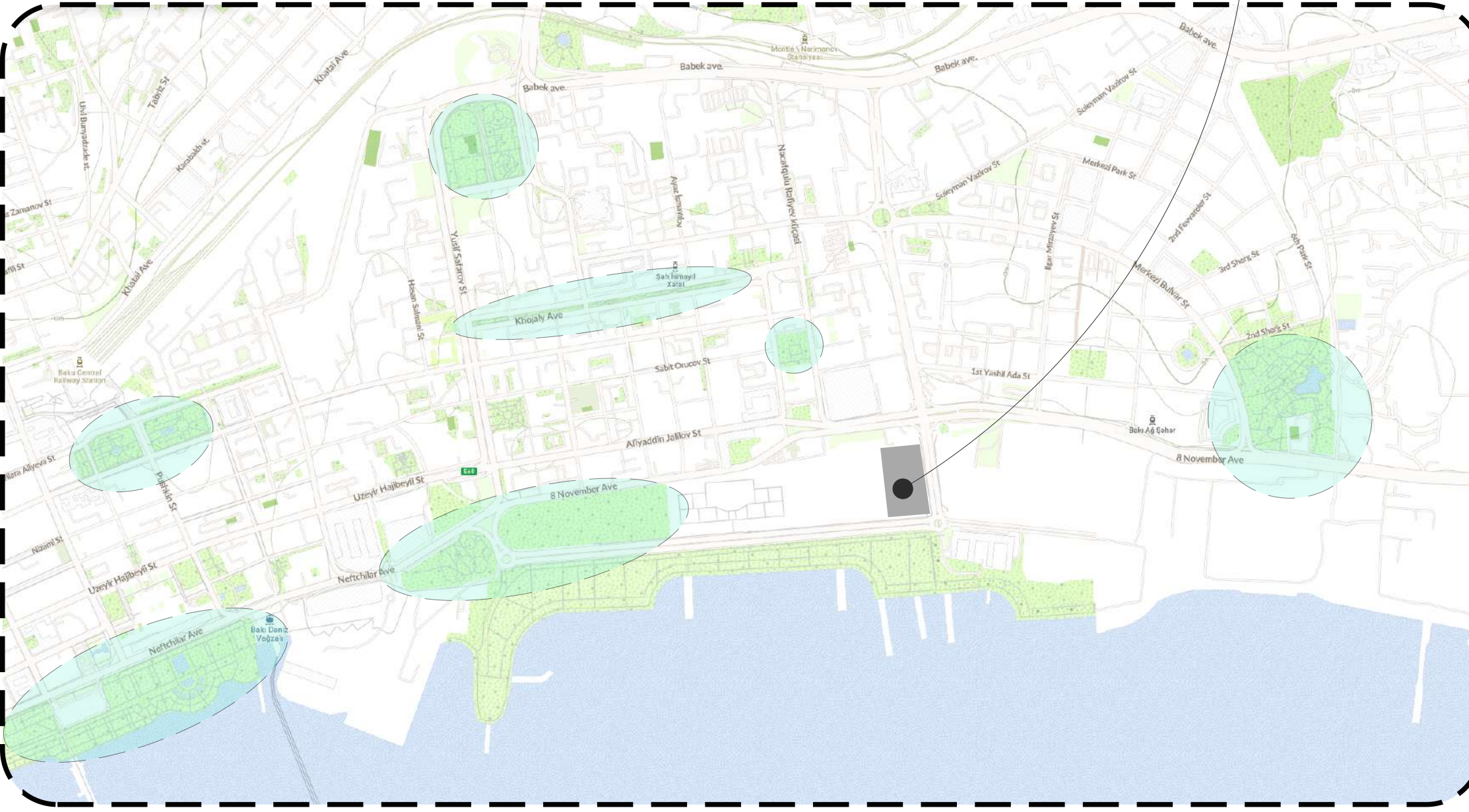
Italian Cypress

These trees are fast-growing, tolerant of urban pollution, and commonly found in streetscapes for providing shade. In our site, these trees are planted on almost all the site providing good shadow and greenery.



Ailanthus

These trees are fast-growing, tolerant of urban pollution, and commonly found in streetscapes for providing shade. These trees are sparsely planted on a few areas on the site. These trees are not ever green meaning they shed there leaves in the autumn and winters.



4. SITE ANALYSIS - Existing Situation



Overview

The site is located in Baku’s White City district, a newly redeveloped area that was previously an industrial zone. It is positioned adjacent to White City Boulevard, offering direct access to the waterfront and scenic views of the Caspian Sea. The surrounding context includes a mix of modern commercial, residential, and business buildings, with ongoing development in nearby plots. The site itself is vacant and prepared for construction, with a flat terrain that makes it highly suitable for large-scale public projects like a museum or cultural center. However, the lack of existing greenery or landscape elements means that strategic urban design will be required to enhance the site’s visual and environmental quality.

From an infrastructure perspective, the site is well-connected with wide roads and pedestrian pathways, ensuring smooth accessibility for visitors. The proximity to business and residential zones means the museum will attract a mix of tourists, professionals, and local residents. Additionally, the open space near the waterfront creates an opportunity for outdoor exhibitions and public engagement spaces. However, being a reclaimed area, the land may require soil treatment and structural assessment before construction, ensuring long-term stability and sustainability.

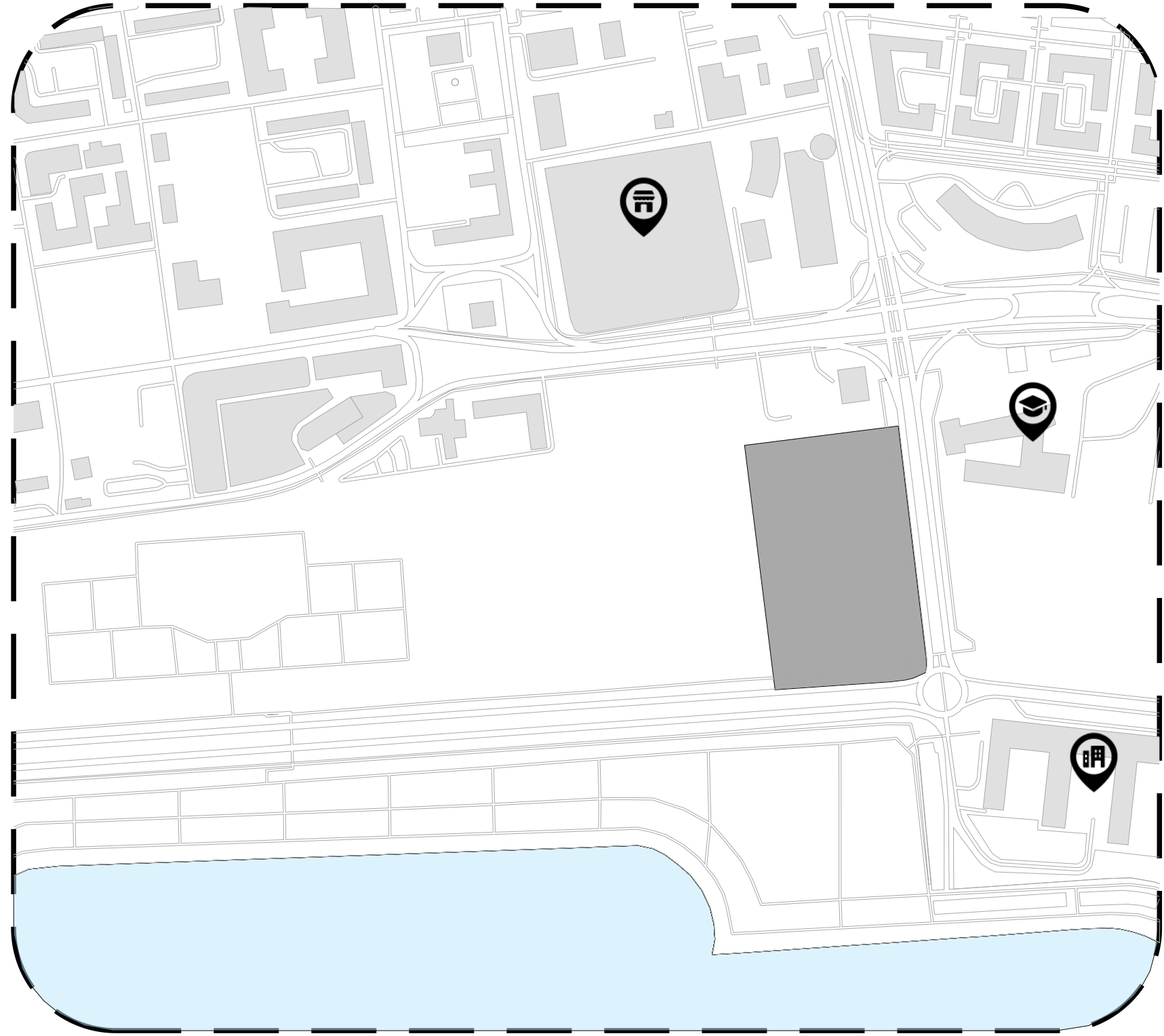
Environmental factors play a significant role in shaping the site’s potential. Being close to the Caspian Sea, the area experiences strong winds, particularly from the north, which must be considered in the building orientation and design of outdoor spaces. The climate consists of hot summers and mild winters, making shading, ventilation, and sustainable cooling solutions essential. Despite these challenges, the prime location and waterfront setting provide a unique opportunity to establish a National Museum of Oil and Gas History and Art Gallery, making it a key cultural and educational landmark in Baku’s evolving urban landscape.



5. SITE ANALYSIS - Land Use



- | | | | |
|-------------|------------|-----------|--------|
| Residential | Commercial | Education | Public |
| Mixed-use | Hospital | Office | Others |

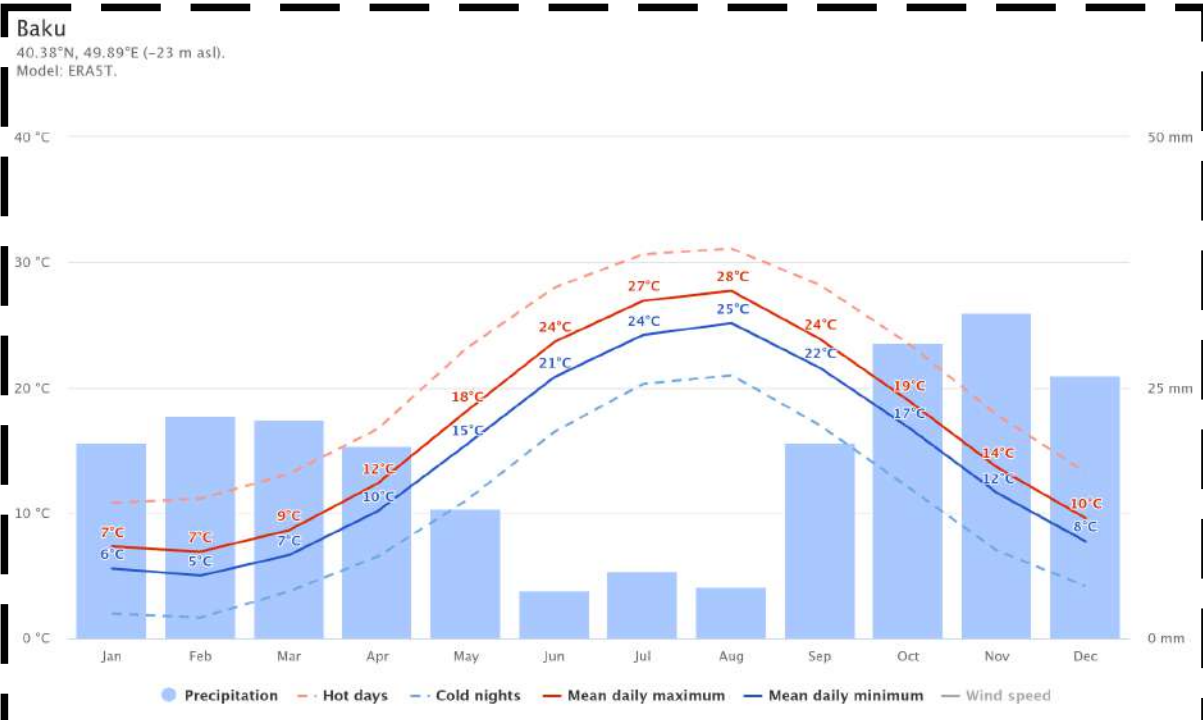


- | | | |
|----------|-------------|-------|
| Medical | Restaurant | Hotel |
| Shopping | Educational | |



- Variation in building height
- | | | |
|-----|------|-----|
| 1-5 | 6-10 | 11+ |
|-----|------|-----|

6. ENVIRONMENTAL ANALYSIS - Sun, Wind, Noise, Shadow & Climate



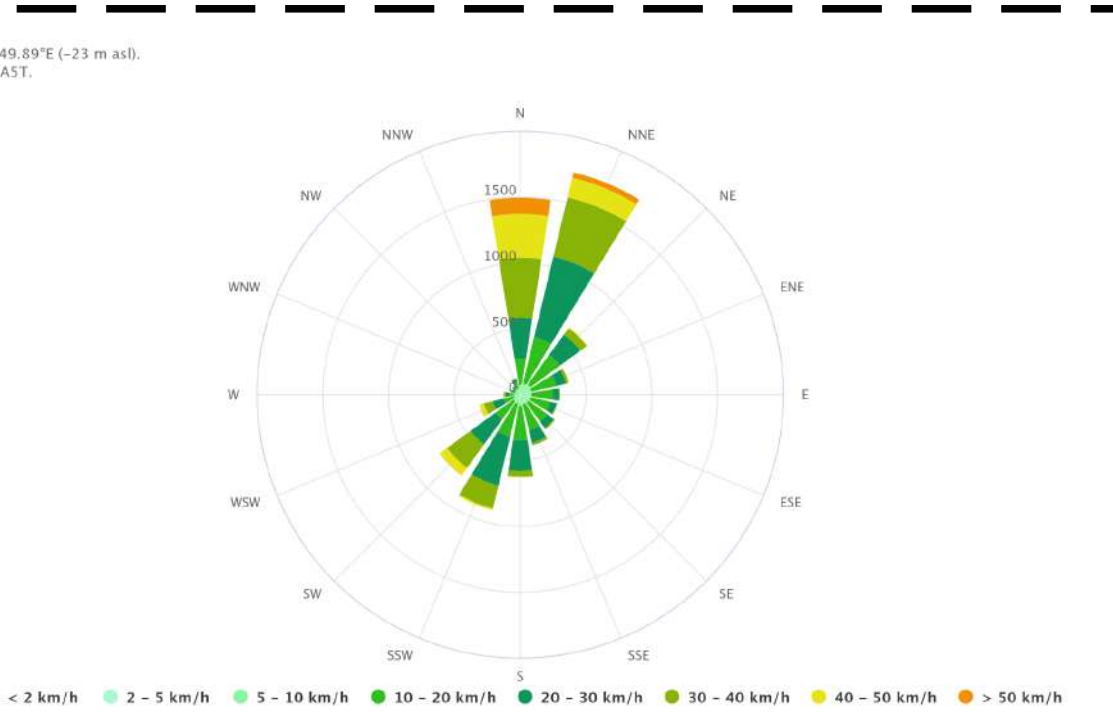
Average Precipitation & Temperature

Overview

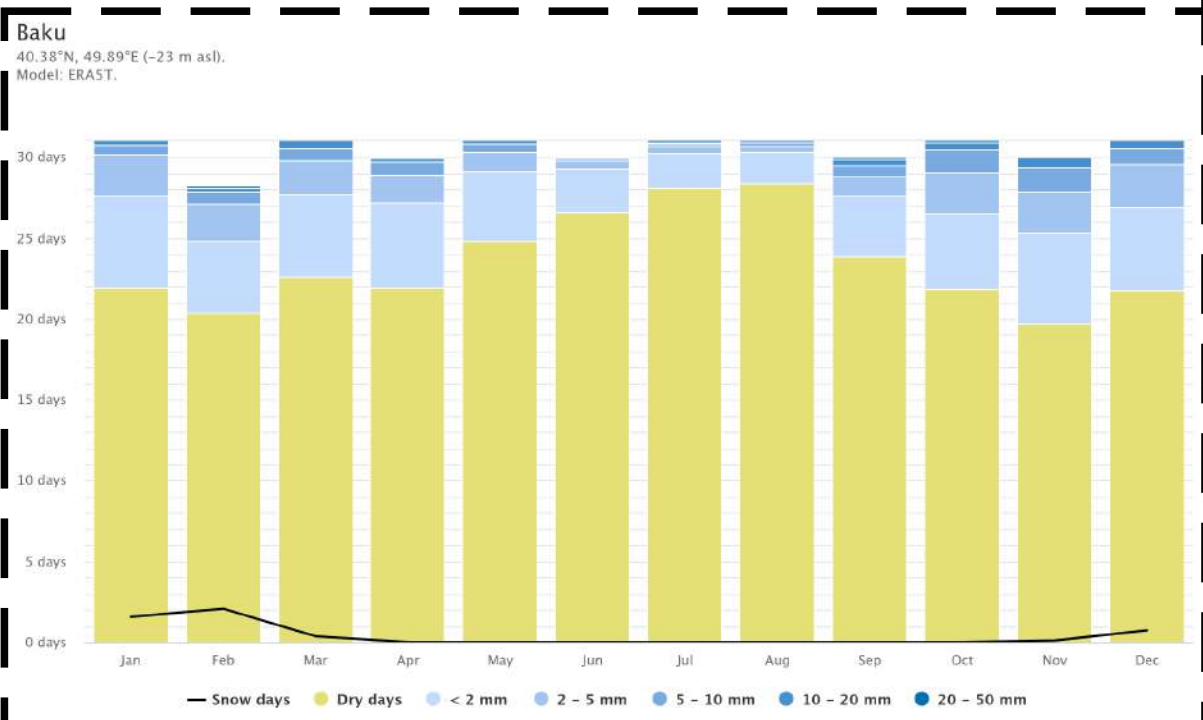
Baku's climate features distinct seasons, with most rainfall occurring in autumn and winter, particularly in November and December, while spring and summer remain dry. The city enjoys abundant sunshine from May to September, ideal for outdoor activities. Summers are warm, reaching up to 30°C, making it perfect for recreational spaces, while winters are cool but mild, with temperatures around 3°C and little snow. Baku's strong, consistent winds, especially in spring and fall, require thoughtful urban design solutions like windbreaks or strategically positioned structures to protect outdoor spaces from gusts.

Effect of Wind

Wind is a significant factor in Baku, known for its consistent and sometimes strong gusts, particularly from the north. The city experiences frequent winds throughout the year, with average speeds of 20 to 30 km/h, peaking during the spring and fall months. For a museum, this requires thoughtful design solutions such as windbreaks, enclosed courtyards, and strategically placed structures to create comfortable outdoor exhibition spaces. Additionally, incorporating aerodynamic building forms and durable materials can help minimize wind impact, ensuring a stable and pleasant environment for both visitors and outdoor installations.



Wind Rose / Direction



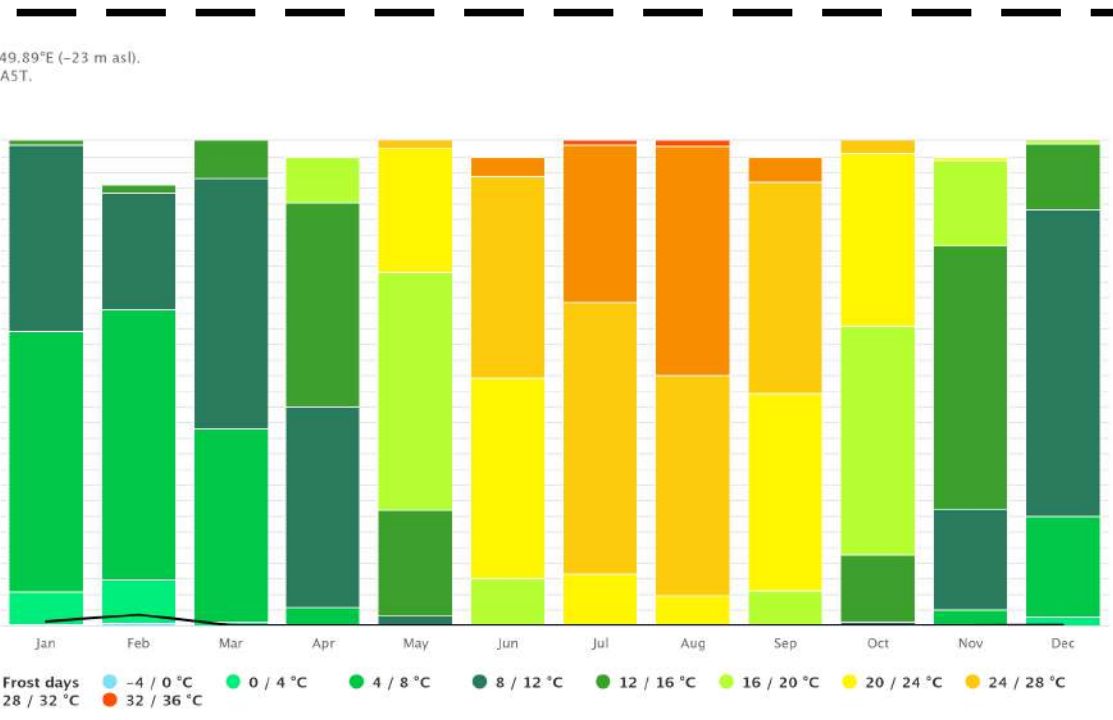
Average Rainfall

Effect of Rainfall

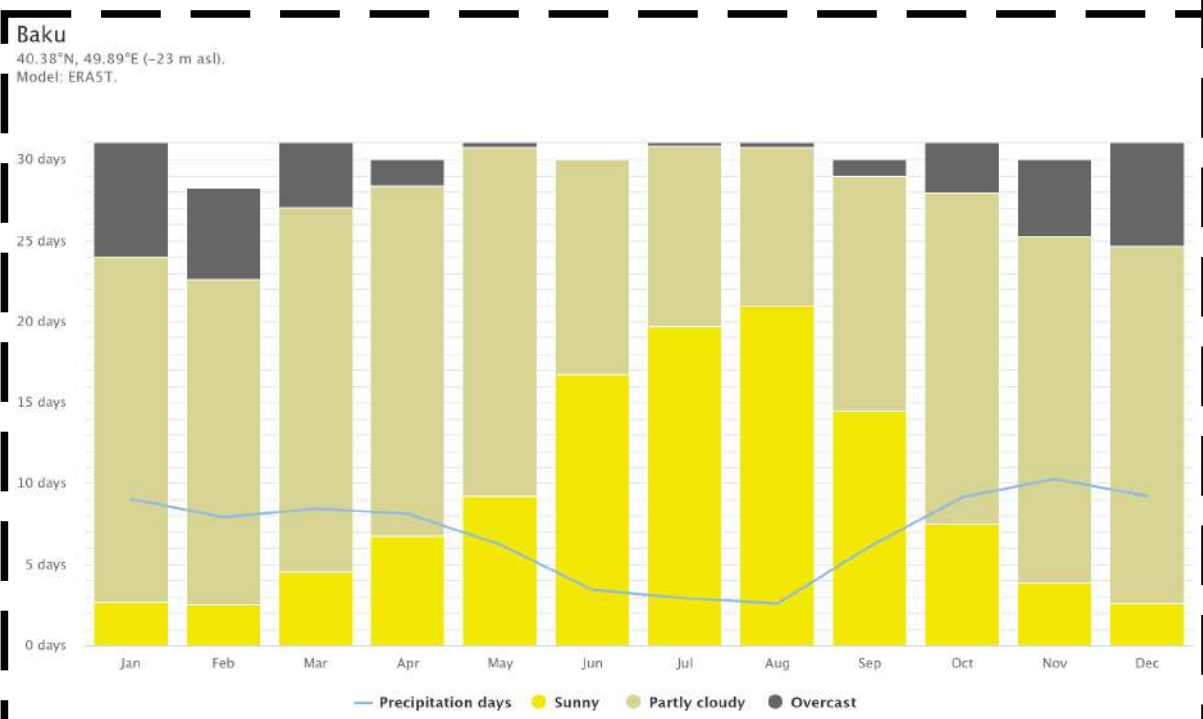
Baku experiences most of its rainfall in the autumn and winter months, particularly in November and December, with precipitation reaching up to 30mm per month. This means that spring and summer are generally dry, providing ideal conditions for outdoor exhibitions, public gatherings, and open-air installations without frequent weather disruptions. In contrast, the wetter months create an opportunity for visitors to explore indoor exhibits, interactive displays, and educational programs that highlight the impact of climate and environmental changes on the region's history and culture.

Effect of Temperature

Baku has warm summers and cool winters, influencing how museum spaces are designed for visitor comfort. During the summer months of June, July, and August, temperatures can rise to 30°C (86°F), making shaded outdoor areas, cooling systems, and well-ventilated indoor spaces essential for a pleasant experience. In winter, from December to February, temperatures drop to around 3°C (37°F), but extreme cold or heavy snowfall is rare. This allows museums to remain accessible year-round, with indoor galleries providing a comfortable retreat during colder months while still allowing for occasional outdoor exhibitions and events.



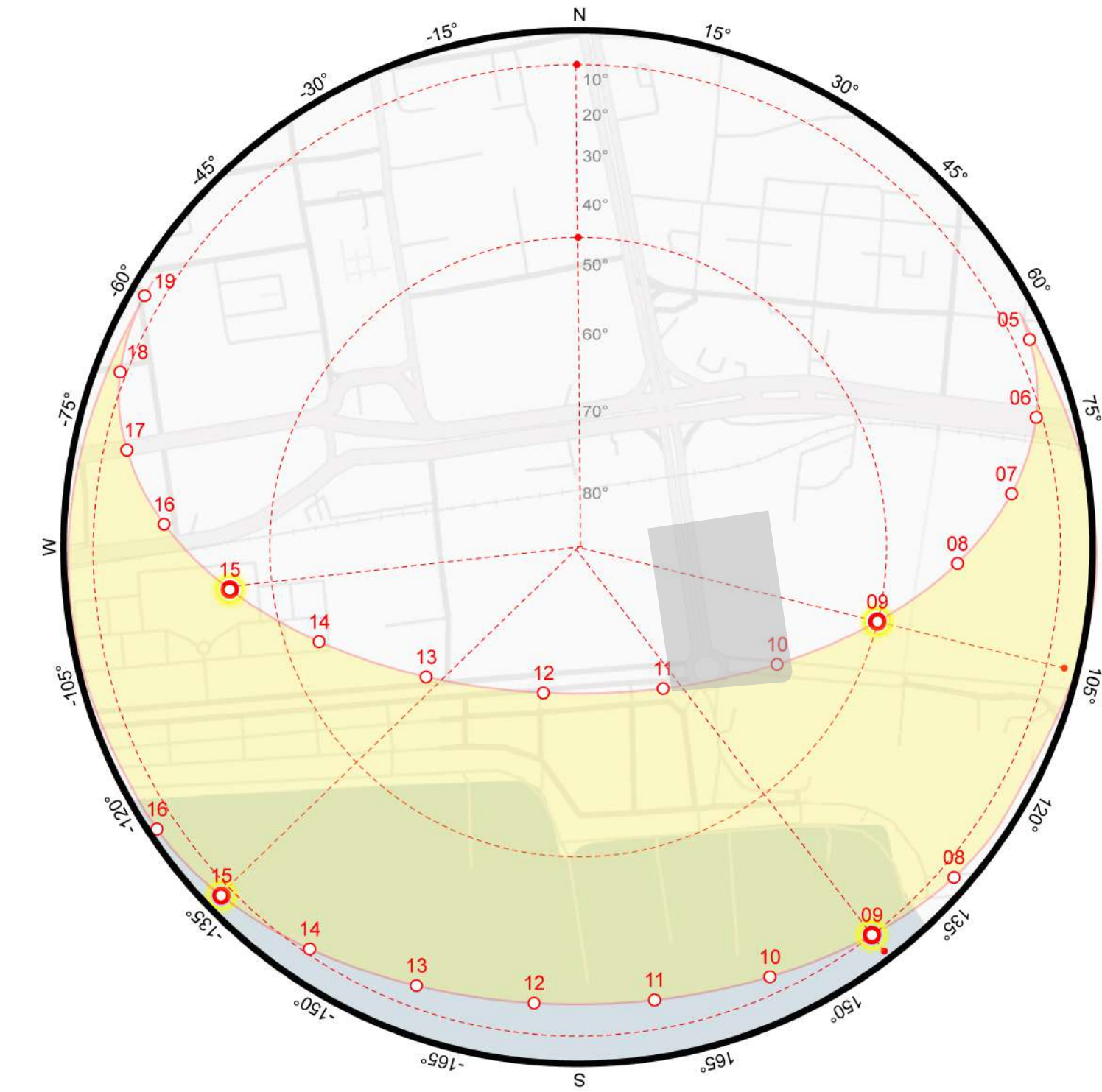
Average Temperatures



Cloudy & Sunny Days

Effect of Sunlight

Baku enjoys abundant sunshine, especially from May to September, with each month experiencing over 20 sunny days. This extended period of sunlight provides excellent natural illumination for museums, enhancing the visitor experience in exhibition halls and reducing the need for artificial lighting. The bright and sunny weather also creates opportunities for outdoor installations, sculpture gardens, and open-air events, allowing visitors to engage with exhibits in a dynamic setting. During winter, while the days are shorter, there is still enough natural light to keep indoor spaces well-lit and inviting, ensuring a comfortable and engaging atmosphere for learning and cultural exploration.



Solar Study

Summer Solstice - June 21
Day Length: 15:03 hrs

9:00 Sun Azimuth: 103.92°
Sun Altitude: 52.00°

15:00 Sun Azimuth: -96.92°
Sun Altitude: 45.50°

Winter Solstice - December 21
Day Length: 9:17 hrs

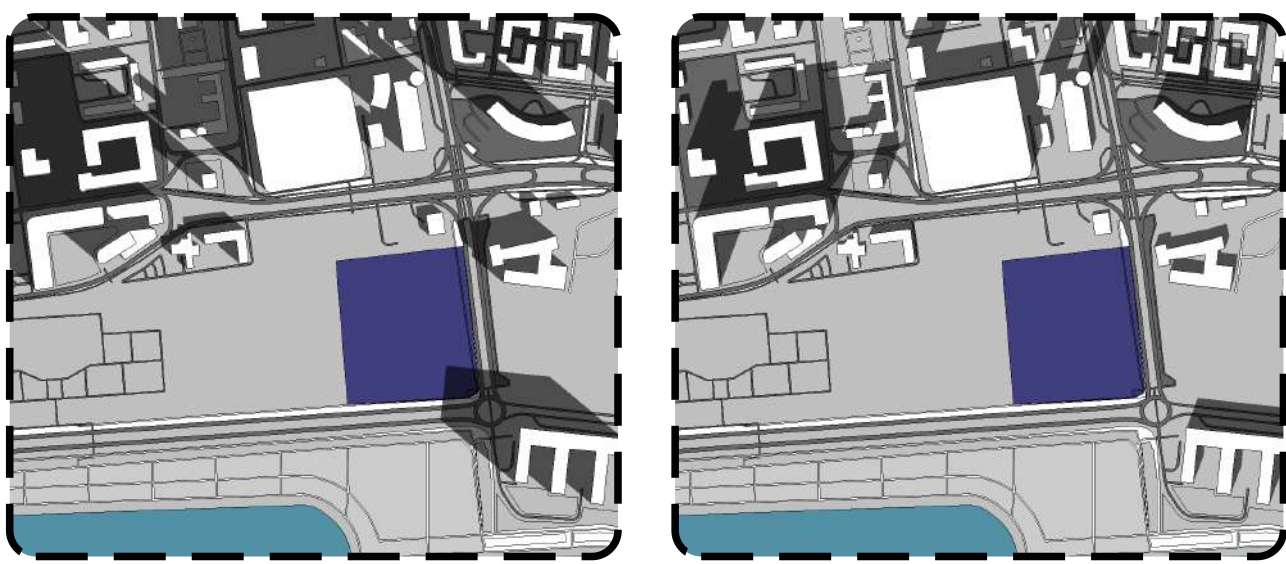
9:00 Sun Azimuth: 142.39°
Altitude: 16.32°

15:00 Sun Azimuth: -134.03°
Sun Altitude: 10.95°

Spring / Fall Equinox
Day Length: 9:17 hrs

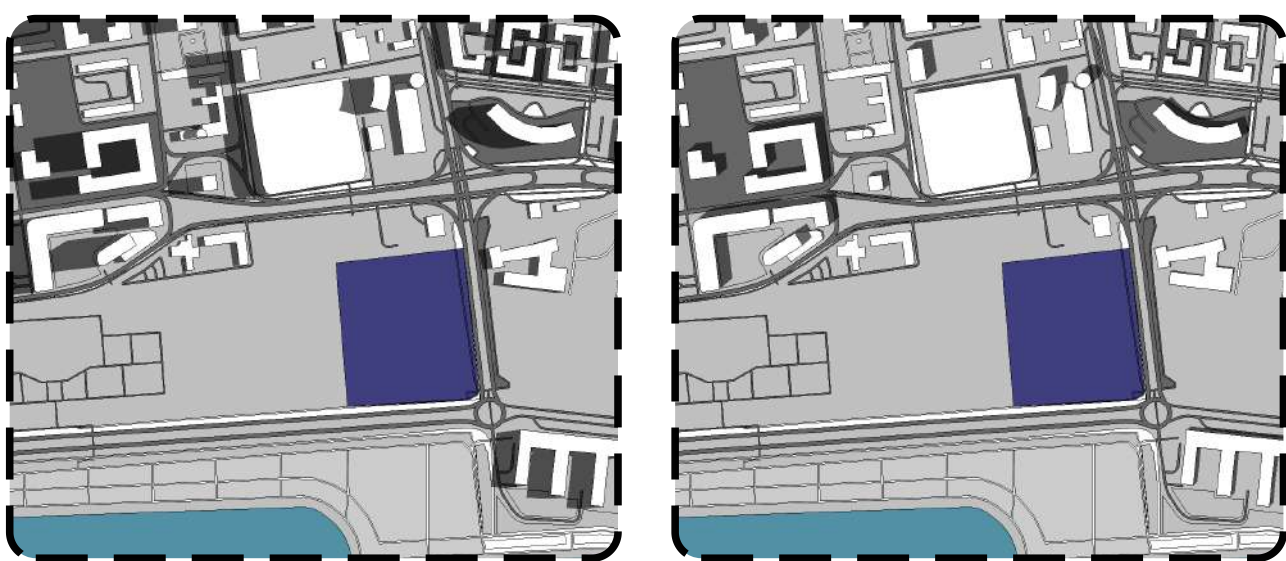
9:00 Sun Azimuth: 128.94°
Altitude: 37.09°

15:00 Sun Azimuth: -116.78°
Sun Altitude: 28.61°



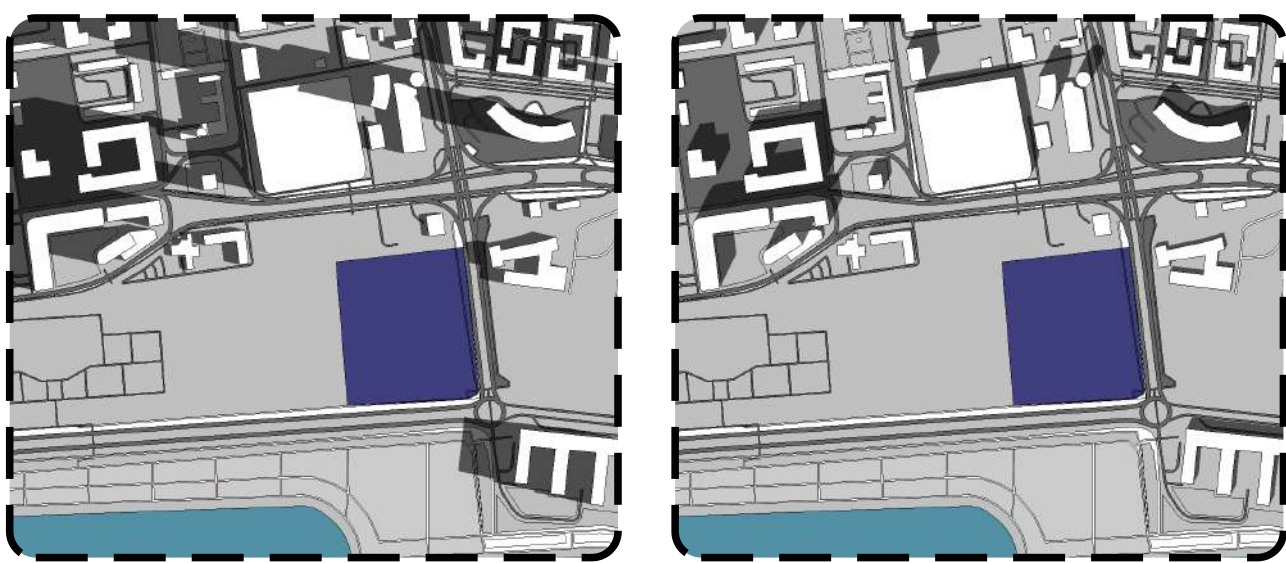
Dec 21 - 9:00 AM

Dec 21 - 3:00 PM



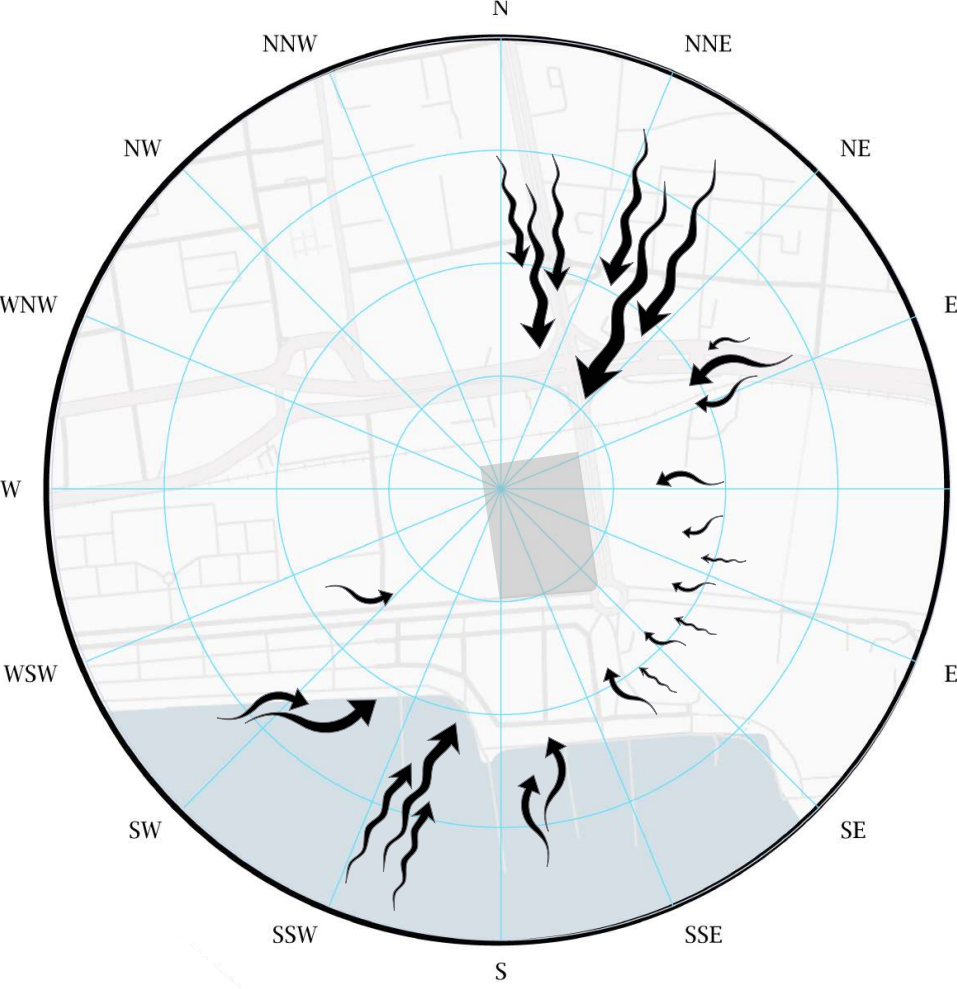
Jun 21 - 9:00 AM

Jun 21 - 3:00 PM



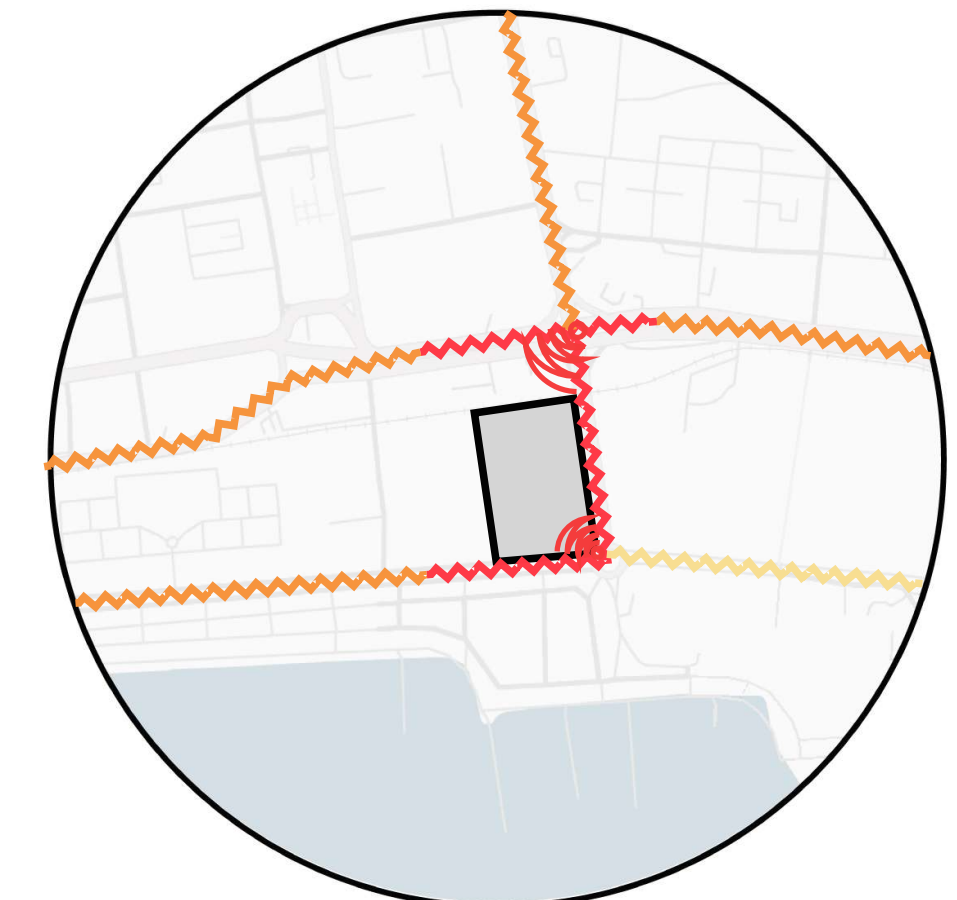
Spring/Fall Equinox - 9:00 AM

Spring/Fall Equinox - 3:00 PM



Wind Direction

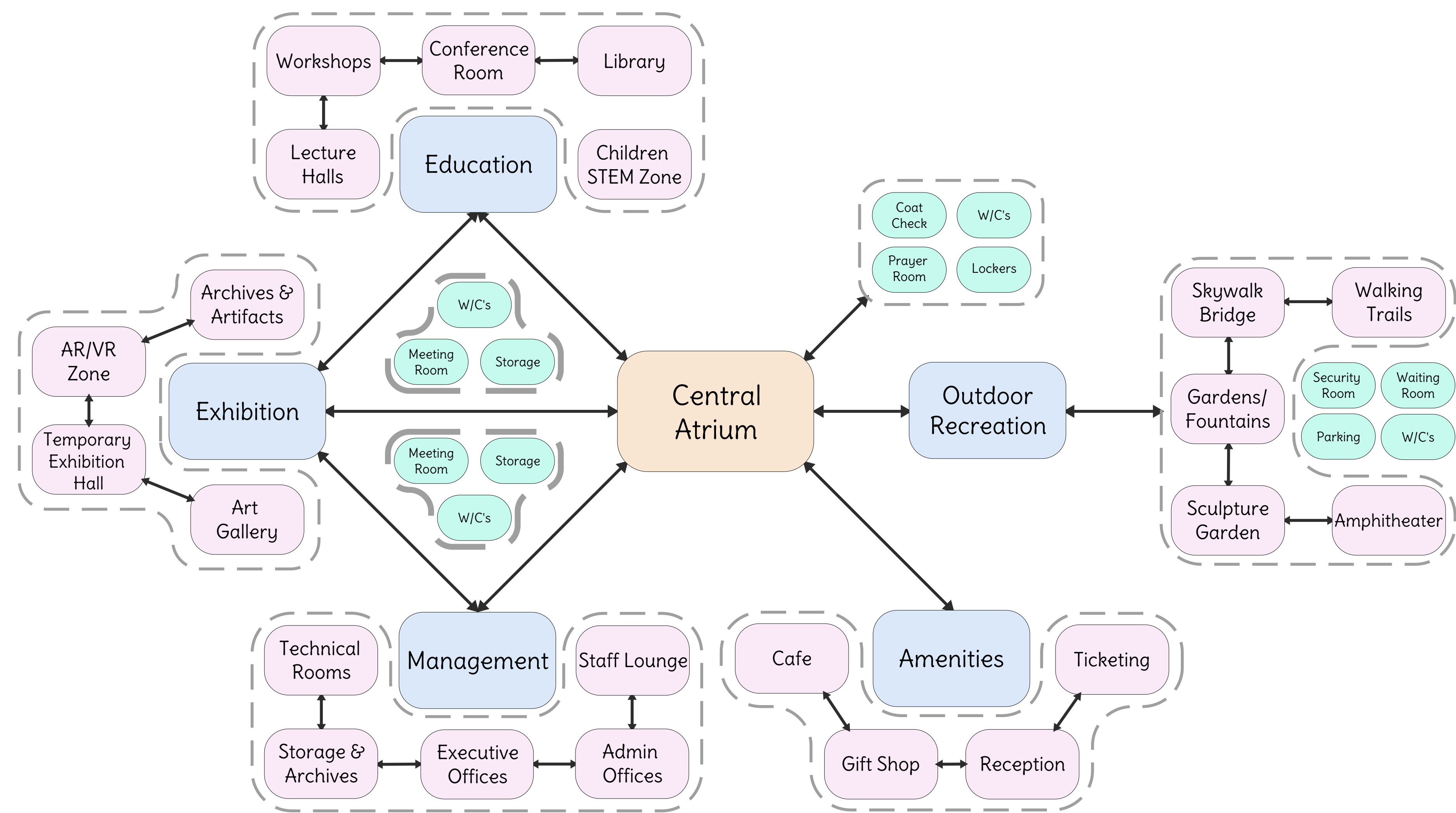
Wind is the most important consideration in the design process as the wind speeds can easily reach more than 50+ kph. The building should be designed in a way that it shields the outdoor areas from these cold northern winds so they can be fully utilized in the winter months also.



Noise Levels

Noise pollution and its sources will be an important consideration for the building design as it is an educational institute. In order to provide the students with a comfortable and noise free environment for studying and development,

7. Bubble Diagram & SWOT Analysis



Strengths

1. There is not lot of overshadowing by neighboring buildings.
2. Proximity to residential buildings and other parks.
3. Rectangular shape and flat terrain makes it easy to design and build on.
4. Easy access is possible as it is well connected with public transport.

Weaknesses

1. Located in a noisy area of the city with major roads on all sides.
2. Air quality is not good during day time due to pollution from vehicles.
3. The renovation work may take a long time due to high surrounding population density.
4. Construction work may be constrained by limited space for machinery and disruption to nearby residents.

Opportunities

1. There are many possibilities of design due to easy shape and terrain.
2. Site already has alot of greenery which can be integrated into the new park.
3. Can be used for making combined parks such as urban and sports together.

Threats

1. Unexpected condition of underground infrastructure - Existing utility lines (water, gas, electricity, etc.) may interfere with construction and lead to additional costs and delays.
2. Public opposition and community dissatisfaction - Certain design changes (e.g., removal of trees, addition of new functions) might not be welcomed by local residents.
3. Negative impact on city traffic - Reconstruction work in the city center may cause congestion, limited access, and traffic disruptions.

8. Space Program

Site Allocation

- Total Site Area: 23,600 sqm
- Outdoor Spaces (Green Areas): 50% = 11,800 sqm
- Setbacks and Parking: 10% = 2,360 sqm
- Remaining Area for Building: 40% = 9,440 sqm

Program Breakdown (Building Area: 9,440 sqm)

1. Exhibition Spaces (45% = ~4,250 sqm)

- Main Exhibition Hall: 1,500 sqm
(History of oil/gas in Azerbaijan, interactive timelines, and artifacts)
- Interactive Technology Zone: 800 sqm
(VR drilling simulations, pipeline mechanics, and renewable energy exhibits)
- Temporary Exhibition Gallery: 600 sqm
(Rotating displays on global energy trends or local innovations)
- Historical Archives & Artifacts: 500 sqm
(Vintage equipment, documents, and photos from Baku's oil boom)
- Outdoor Exhibits (Partial Covered): 850 sqm
(Integrated into green areas: historic rigs, pipelines, and sculptures)

2. Educational Spaces (20% = ~1,890 sqm)

- Auditorium (200 seats): 400 sqm
- Classrooms/Workshops: 600 sqm
- Children's STEM Zone: 400 sqm
(Hands-on experiments and games about energy science)
- Research Library: 300 sqm
(Digital archives, rare books, and study areas)
- Lecture Halls: 190 sqm

3. Visitor Amenities (15% = ~1,415 sqm)

- Lobby & Ticketing: 400 sqm
- Café/Restaurant: 500 sqm
- Gift Shop: 150 sqm
- Prayer Room: 50 sqm
- Restrooms/Lockers: 315 sqm

4. Administrative & Operations (10% = ~945 sqm)

- Staff Offices: 500 sqm
- Security/Maintenance: 200 sqm
- Storage/Archives: 200 sqm
- IT/Server Room: 45 sqm

5. Special Features (10% = ~945 sqm)

- Central Atrium with Kinetic Art: 500 sqm
(A sculptural centerpiece symbolizing fire or energy flow)
- Skywalk Observation Deck: 300 sqm
(Overlooking outdoor exhibits and Baku's skyline)
- VIP Lounge: 145 sqm

Outdoor Spaces (11,800 sqm)

- Sculpture Garden: 2,000 sqm
(Artworks inspired by oil derricks, flames, and Caspian Sea motifs)
- Amphitheater: 800 sqm
(Open-air events and film screenings)
- Walking Trails: 5,000 sqm
(Landscaped paths with interpretive signs about Azerbaijan's oil history)
- Green Zones: 4,000 sqm
(Native plants, shaded seating, and water features)

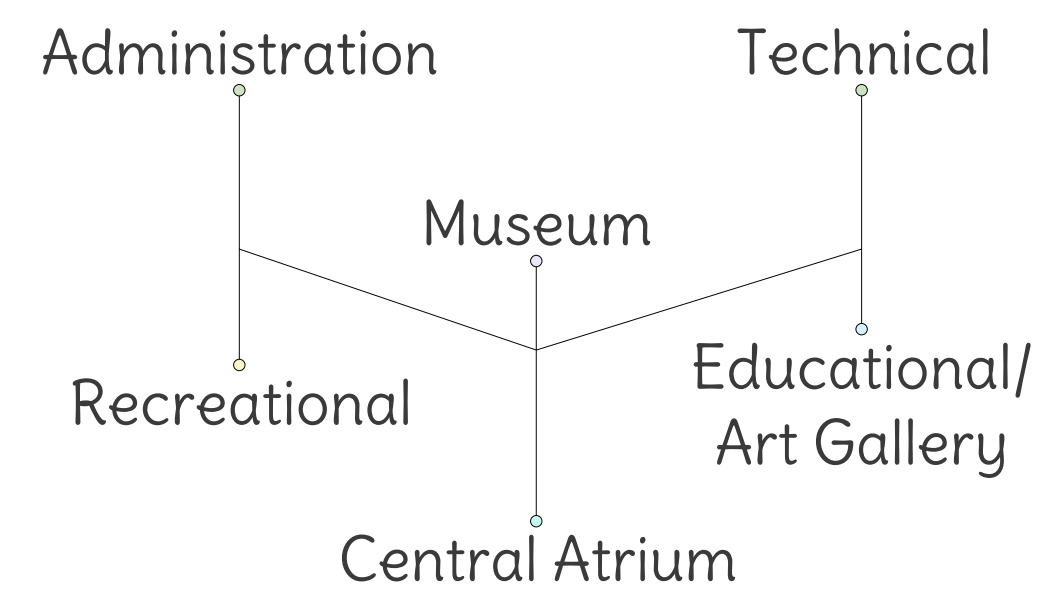
Parking & Setbacks (2,360 sqm)

- Visitor Parking: 2,000 sqm (~60 cars)
- Setbacks (Landscaped Buffer Zones): 360 sqm

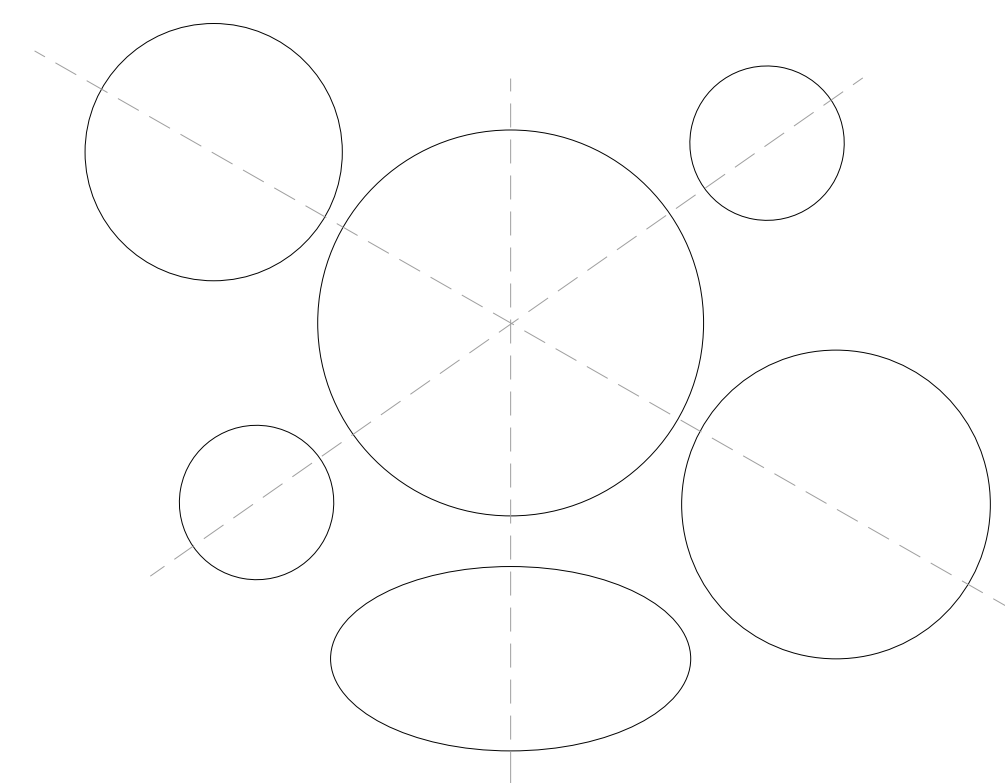
Design Themes

1. Symbolic Architecture:
 - Use **curved, flame-like façades** with amber glass or metallic panels to evoke fire/oil.
 - A **spiral or vortex-shaped roof** to mimic gas movement.
2. Material Palette:
 - **Black granite** (symbolizing oil) + **reflective steel** (gas) + **terracotta accents** (fire).
3. Sustainability:
 - Solar panels on rooftops, rainwater harvesting for green areas, and geothermal heating/cooling.

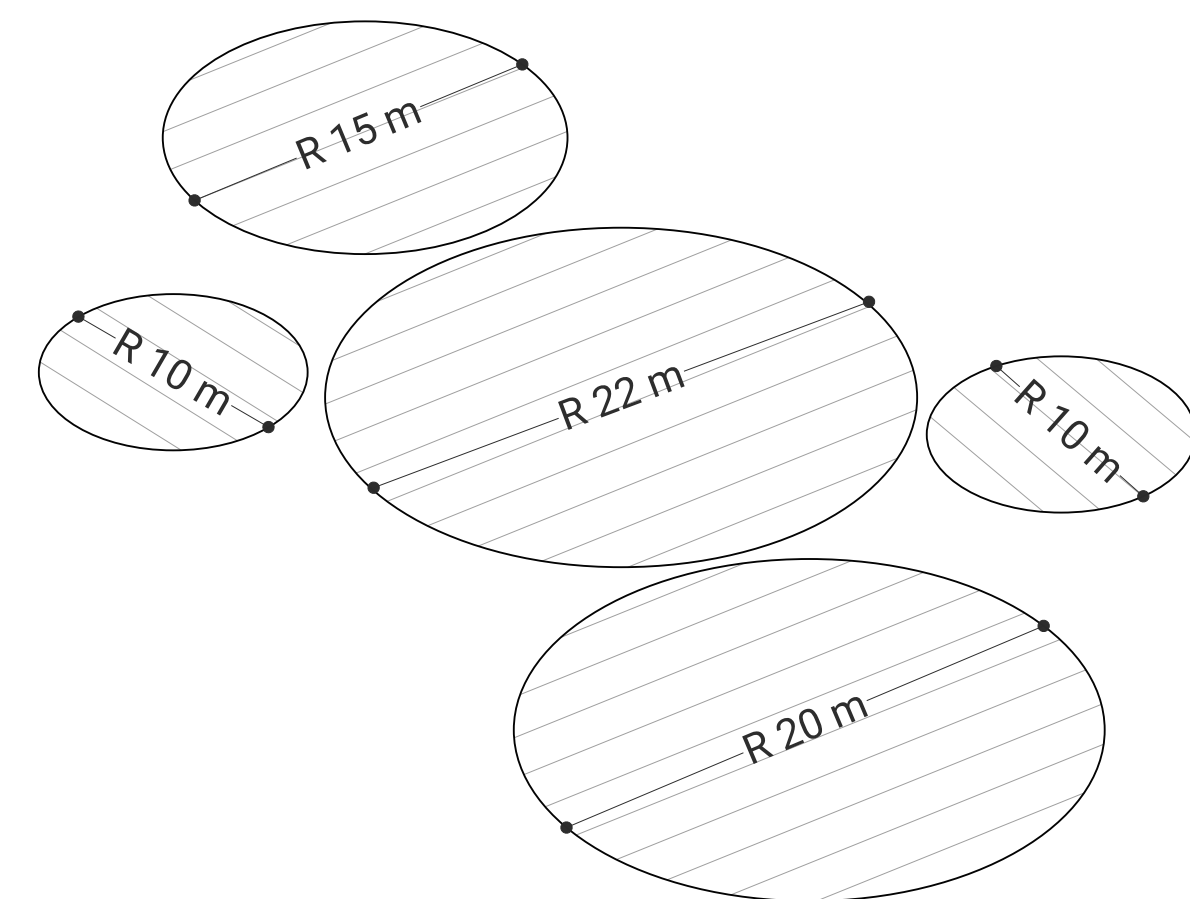
CONCEPT - Building Form & Spatial Distribution



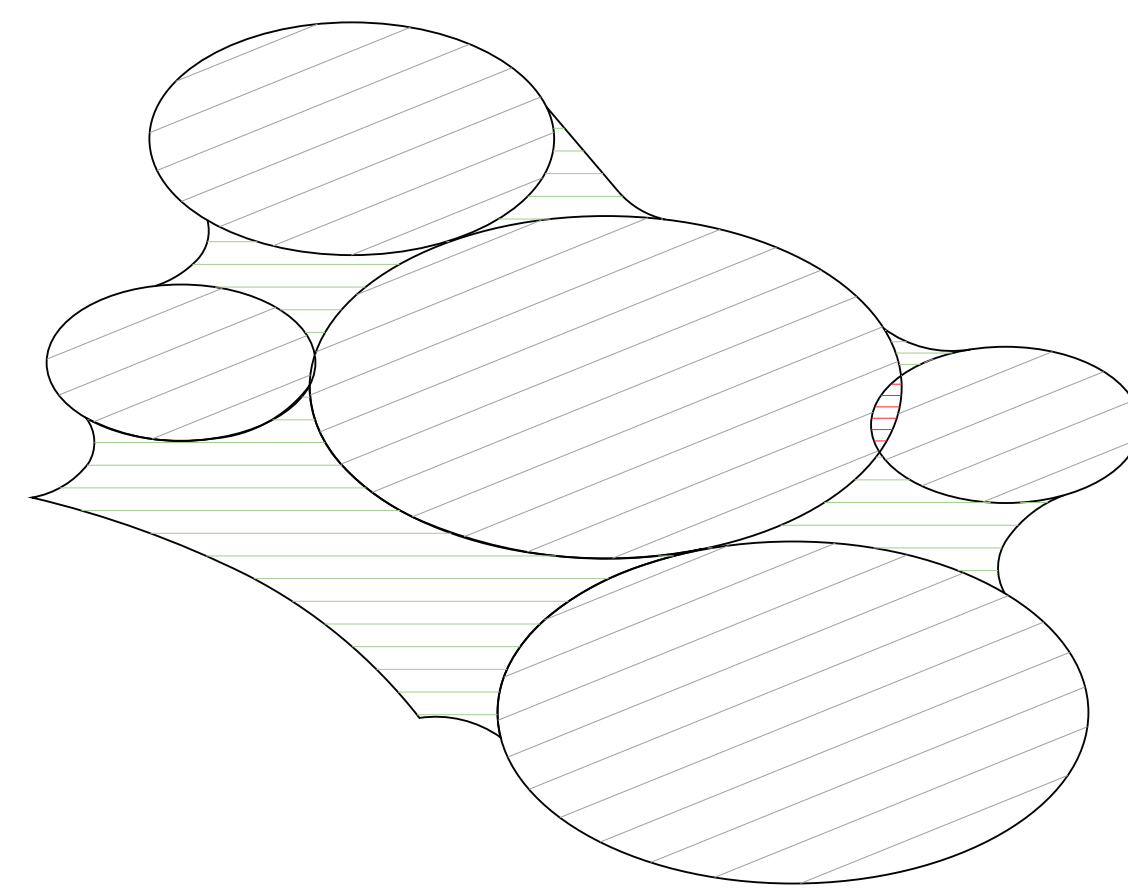
1. Functional zoning based on user needs: Central Atrium connects major zones—Museum, Gallery, Administration, Technical, and Recreational.



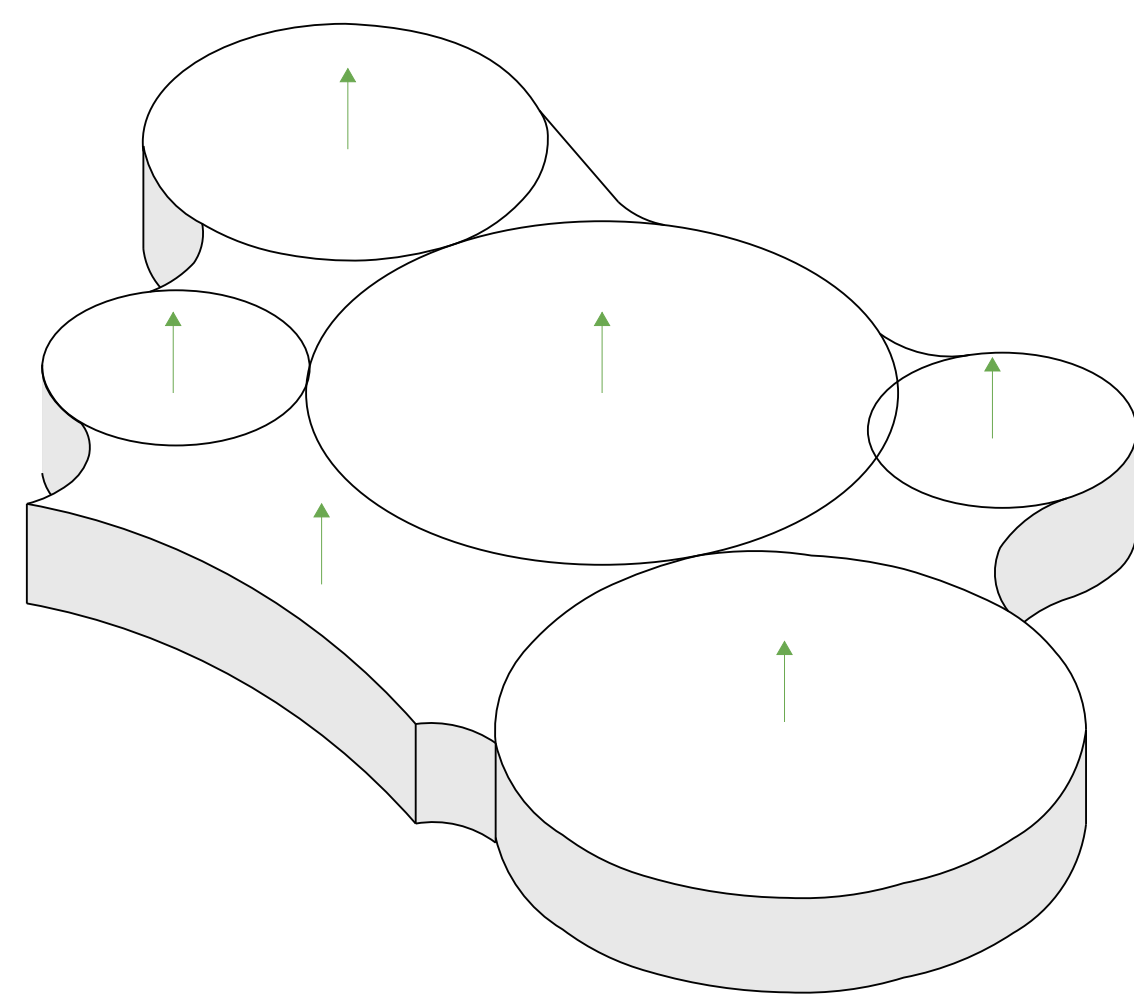
2. Conceptual layout developed using circular modules around a central axis for balance and fluid circulation



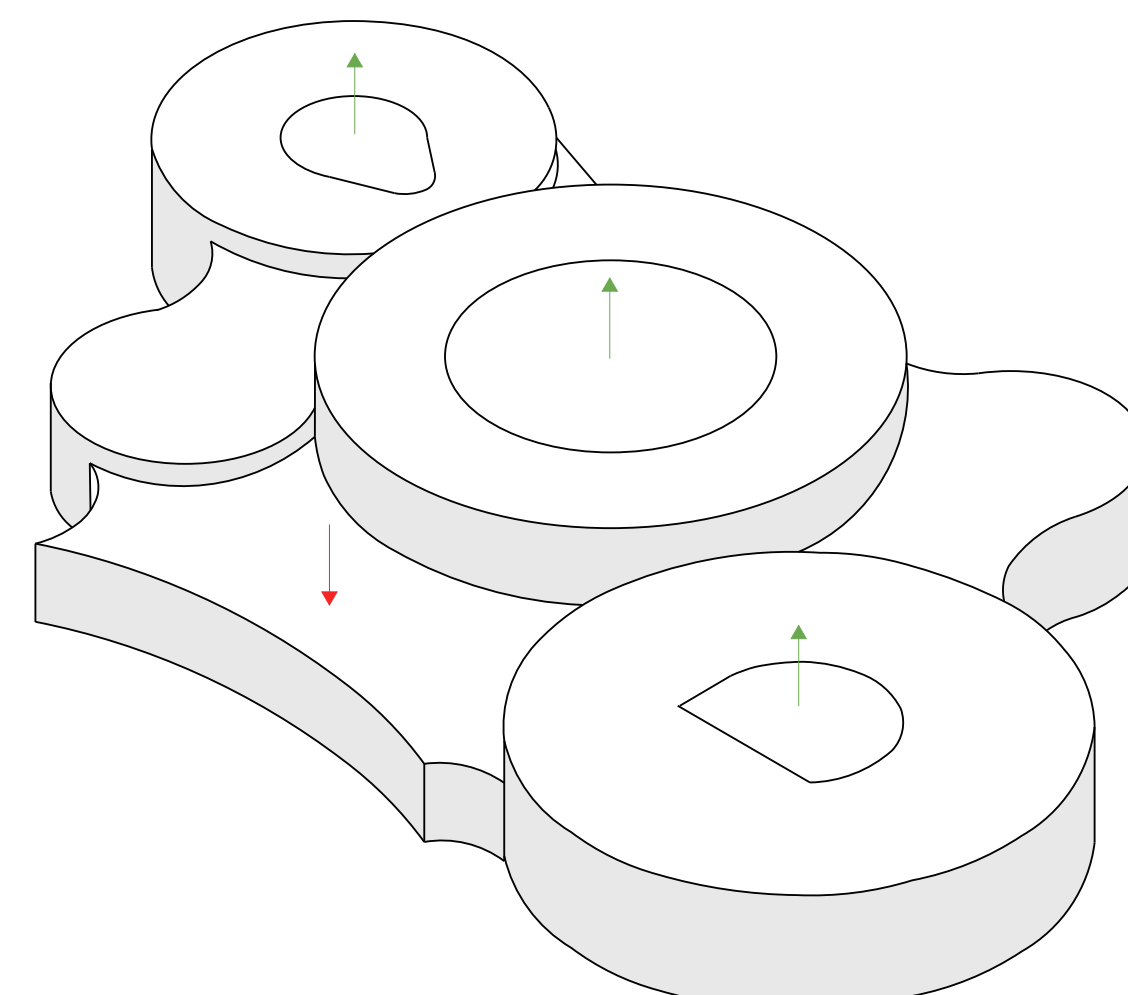
3. Area distribution for each functional zone determined by program requirements.



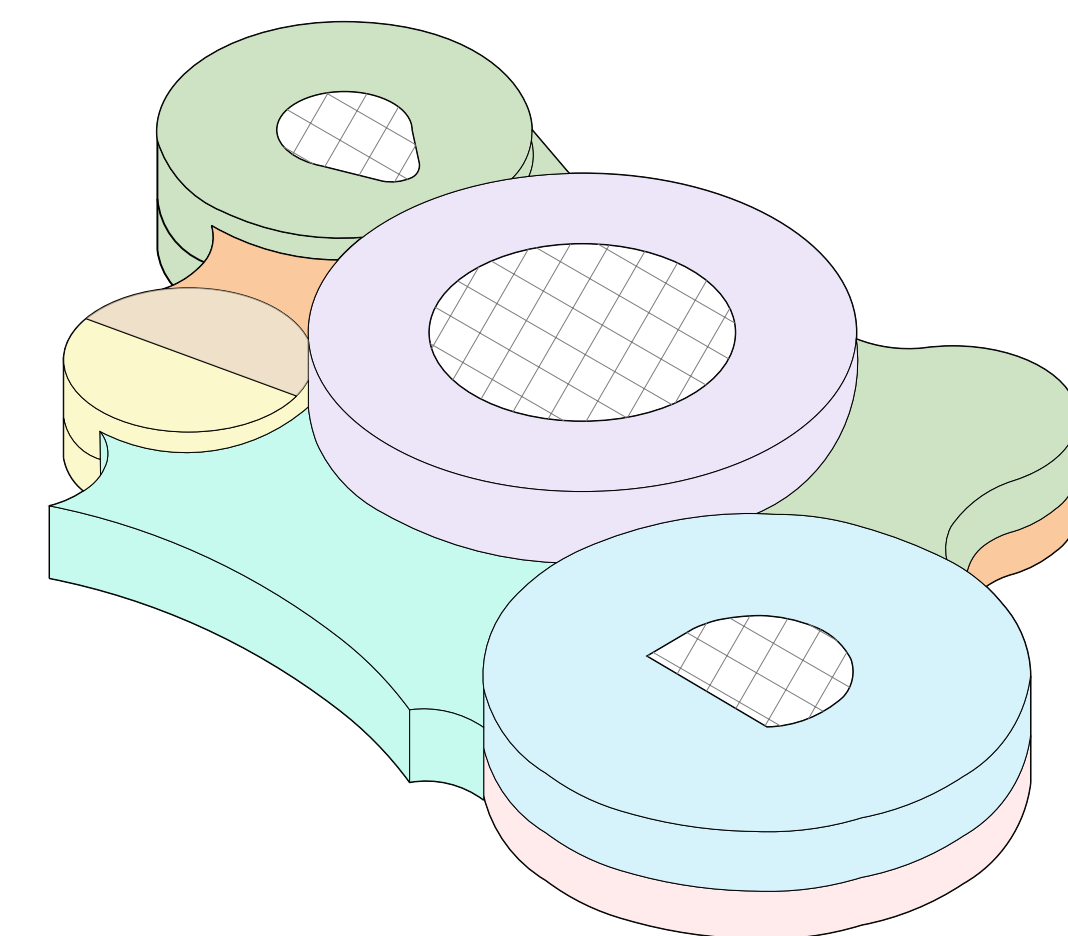
4. Creating level changes and stepped connections for visual and spatial interest.



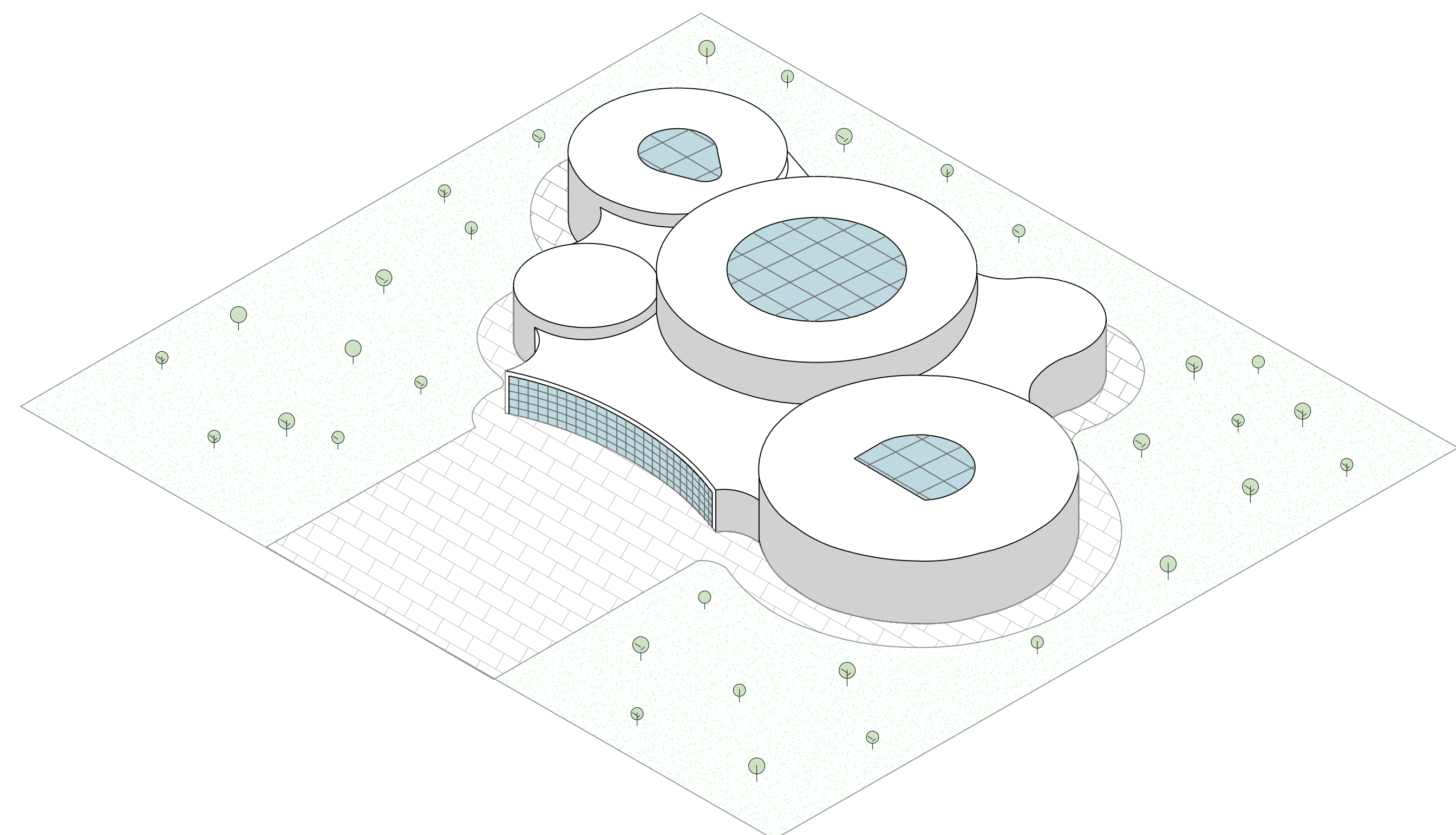
5. Initial massing generated by extruding each programmatic circle to a uniform height based on its spatial requirement.



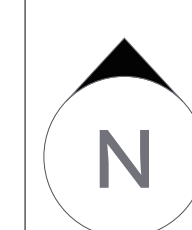
6. Volumes are further adjusted to different heights—introducing vertical variation that reflects the hierarchy and function of each zone.



7. Final 3D composition with color-coded functions: illustrating the spatial organization and relationship between museum, gallery, admin, services, and recreational areas.



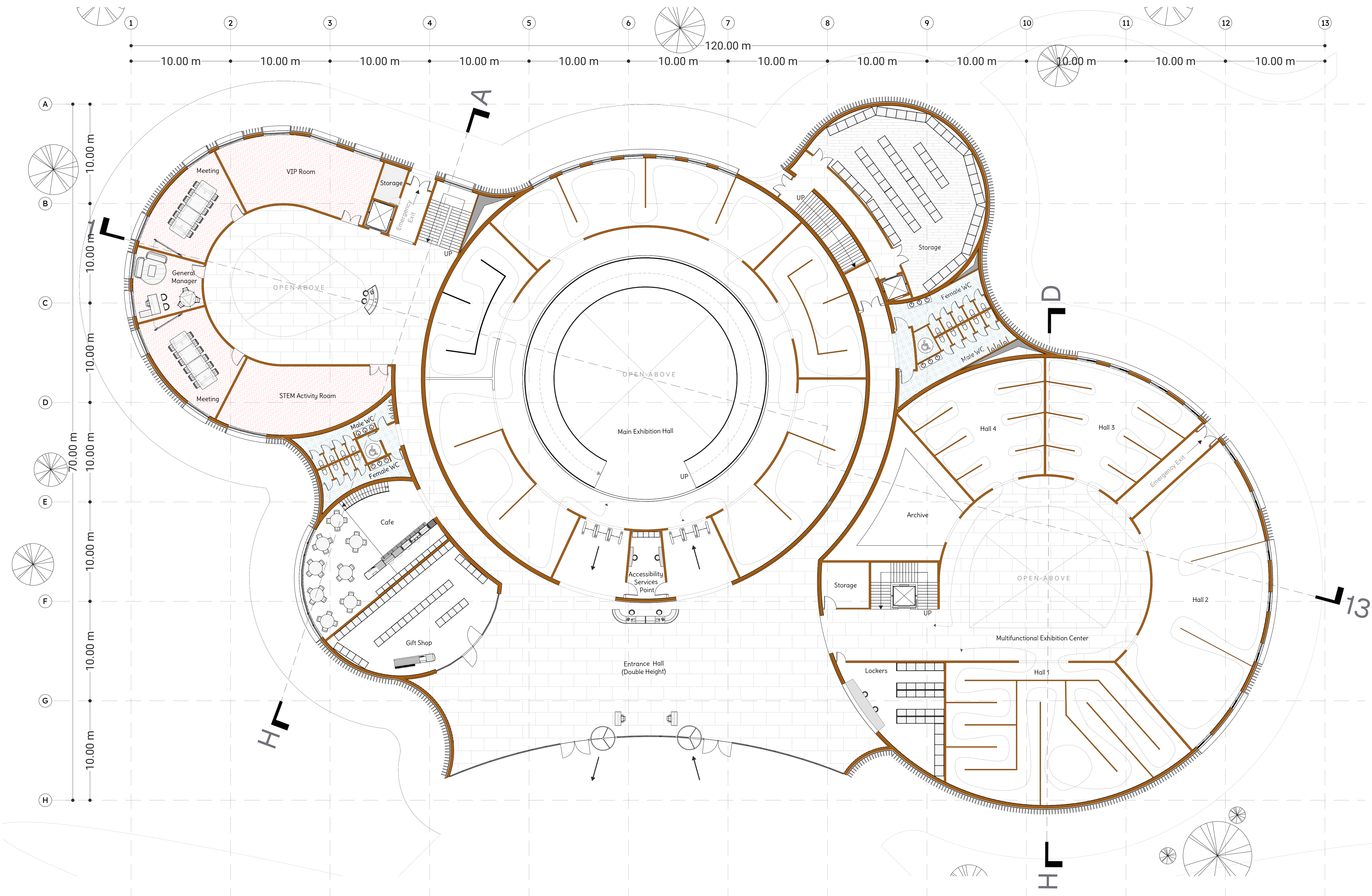
8. Final mass integrated into landscaped site, emphasizing pedestrian approach and soft natural surroundings.



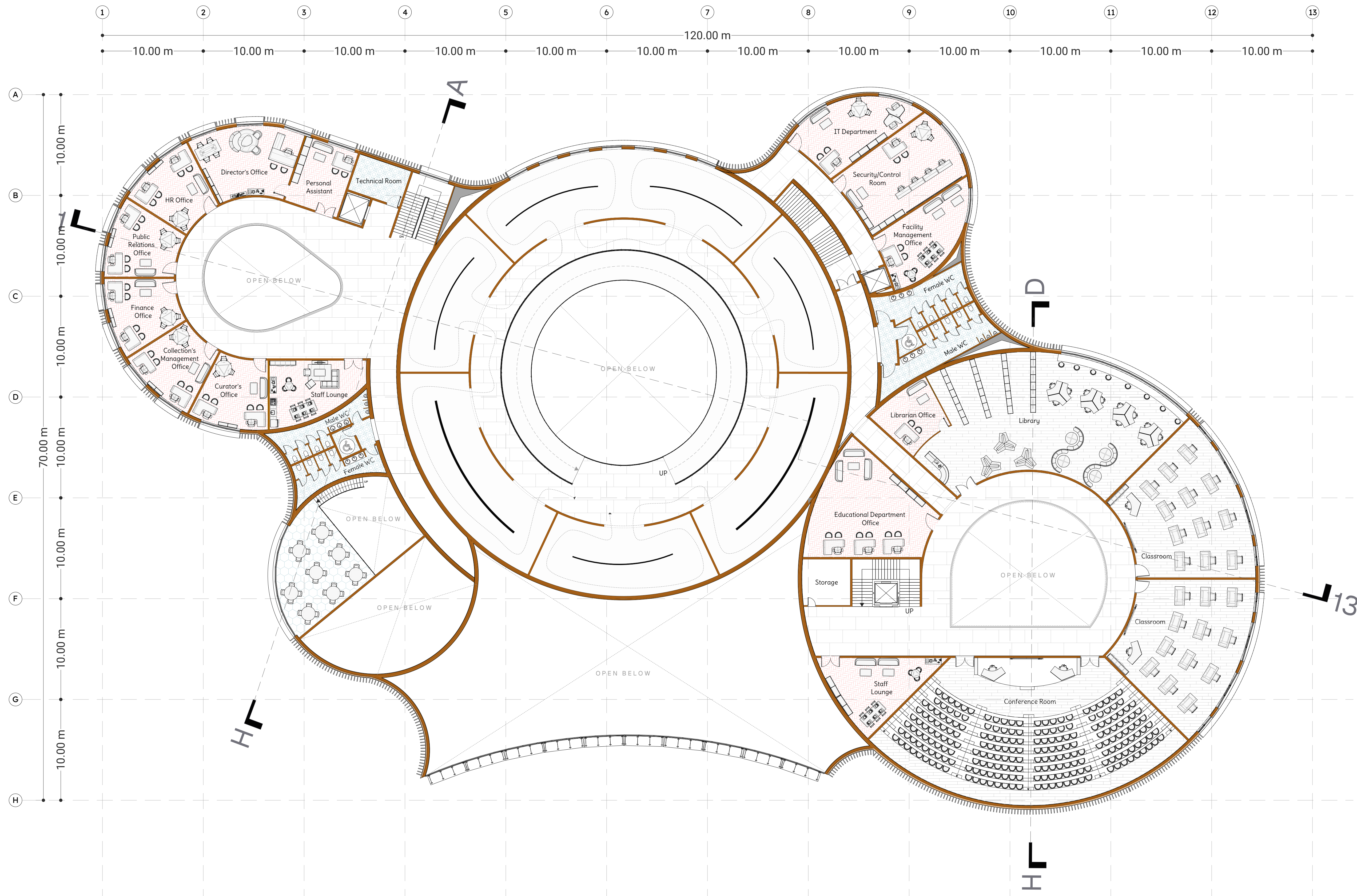
Scale
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ARCHITECTURAL DESIGN - 4
Teacher: MSc Ardit Elezi
Student: Mouiz Ul Hassan

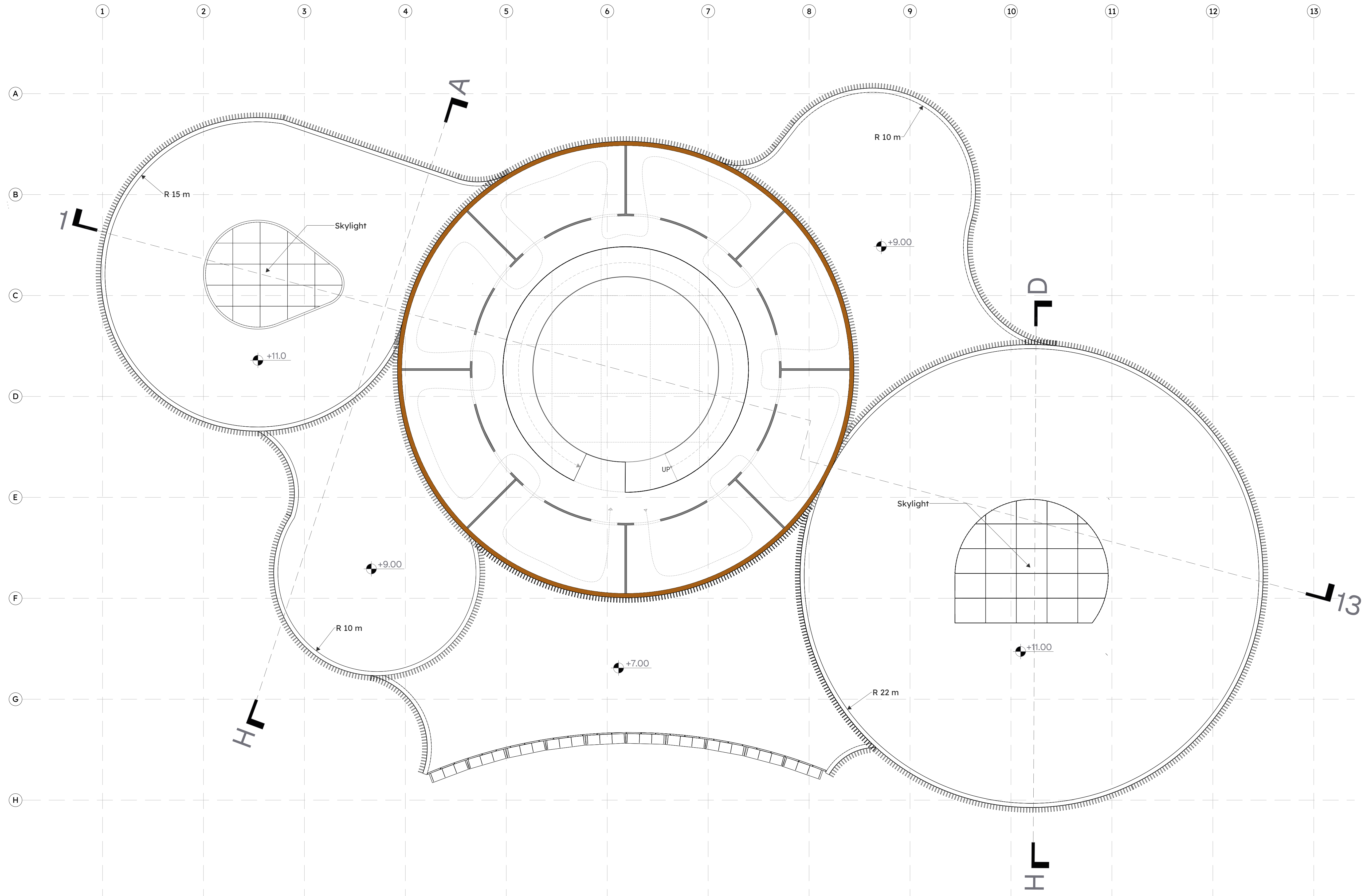
FLOORPLANS - First Floor



FLOORPLANS - Second Floor

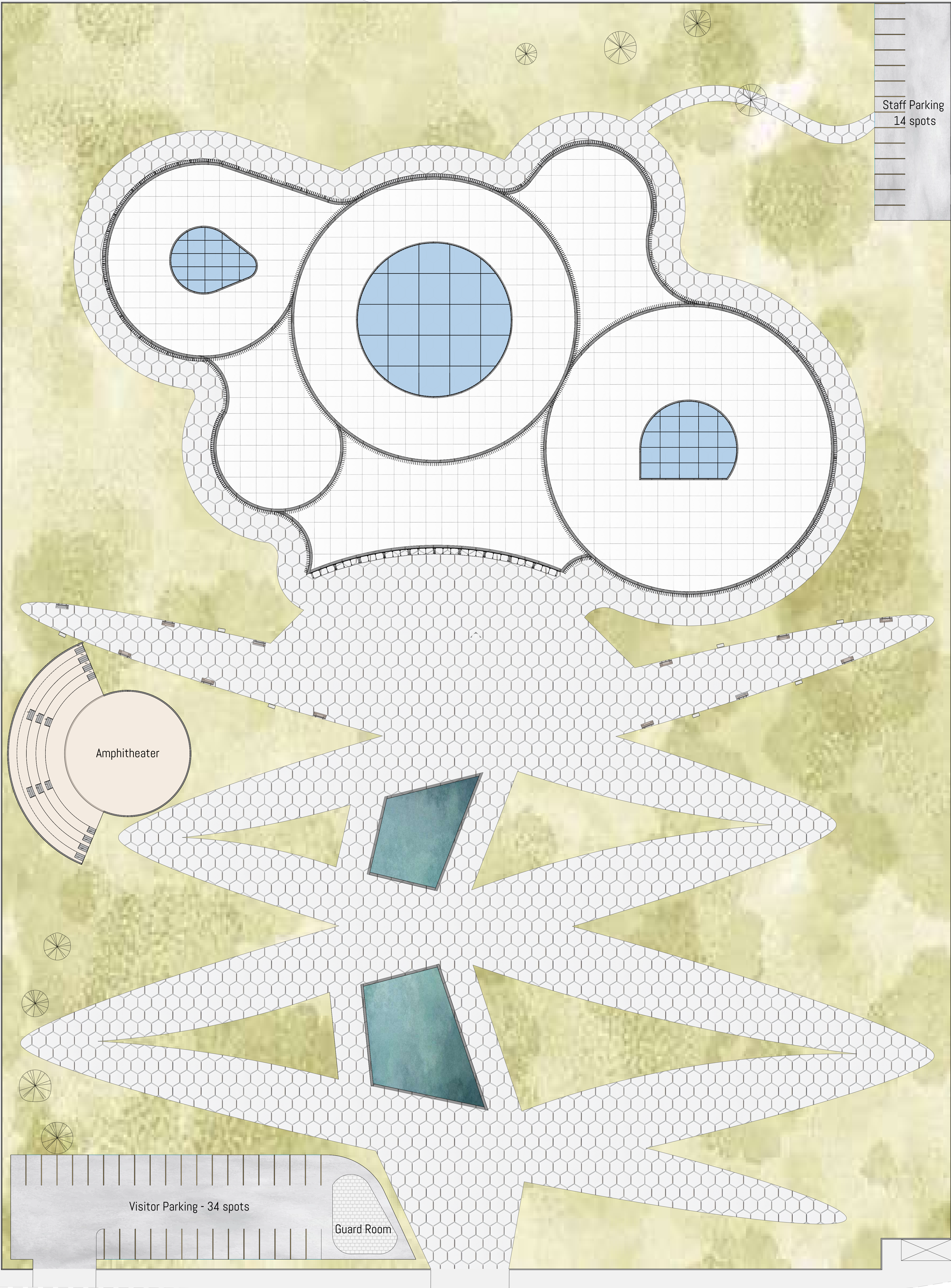


FLOORPLANS - Third Floor



Scale
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ARCHITECTURAL DESIGN - 4
Teacher: MSc Ardit Elezi
Student: Mouiz Ul Hassan



ELEVATIONS | 1 - 13 & 13 - 1



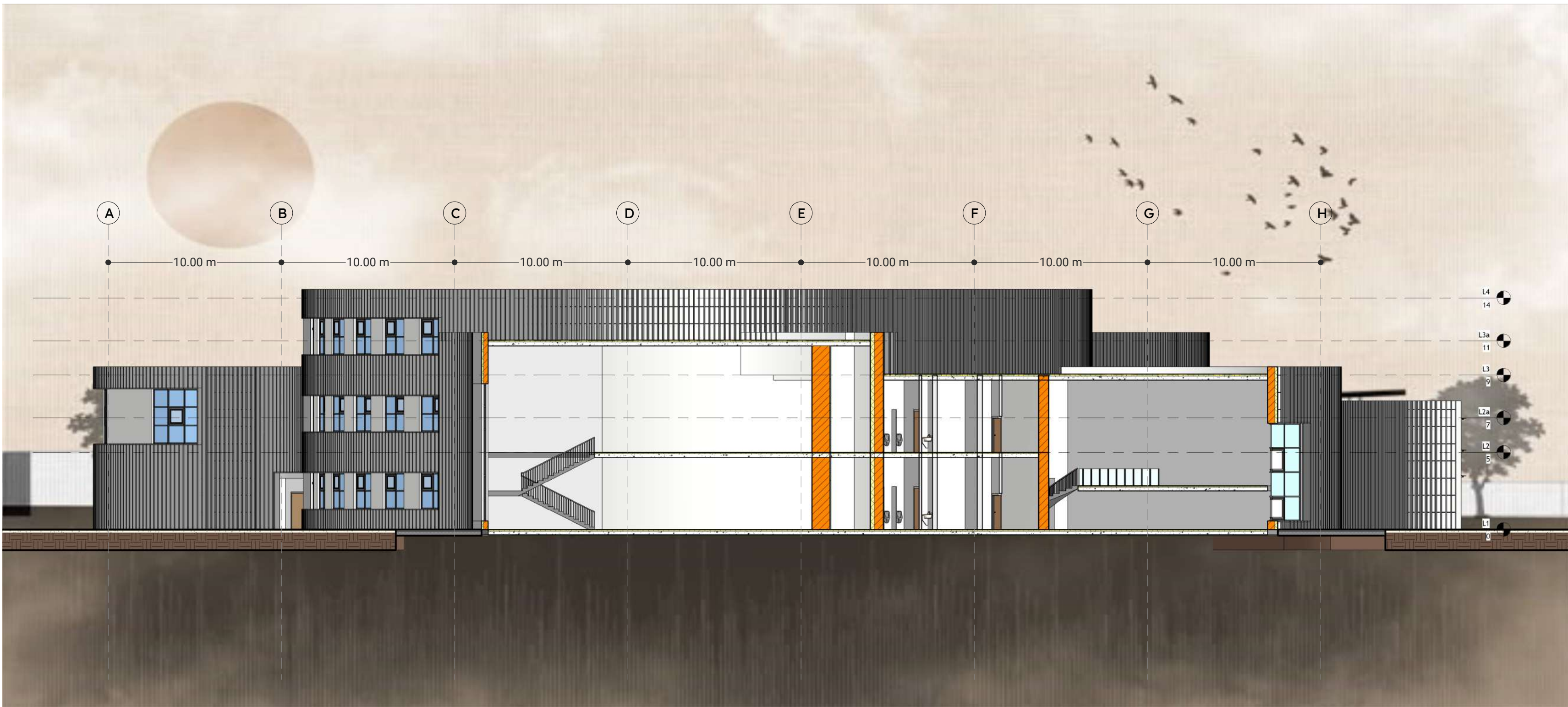
ELEVATIONS | A - H & H - A



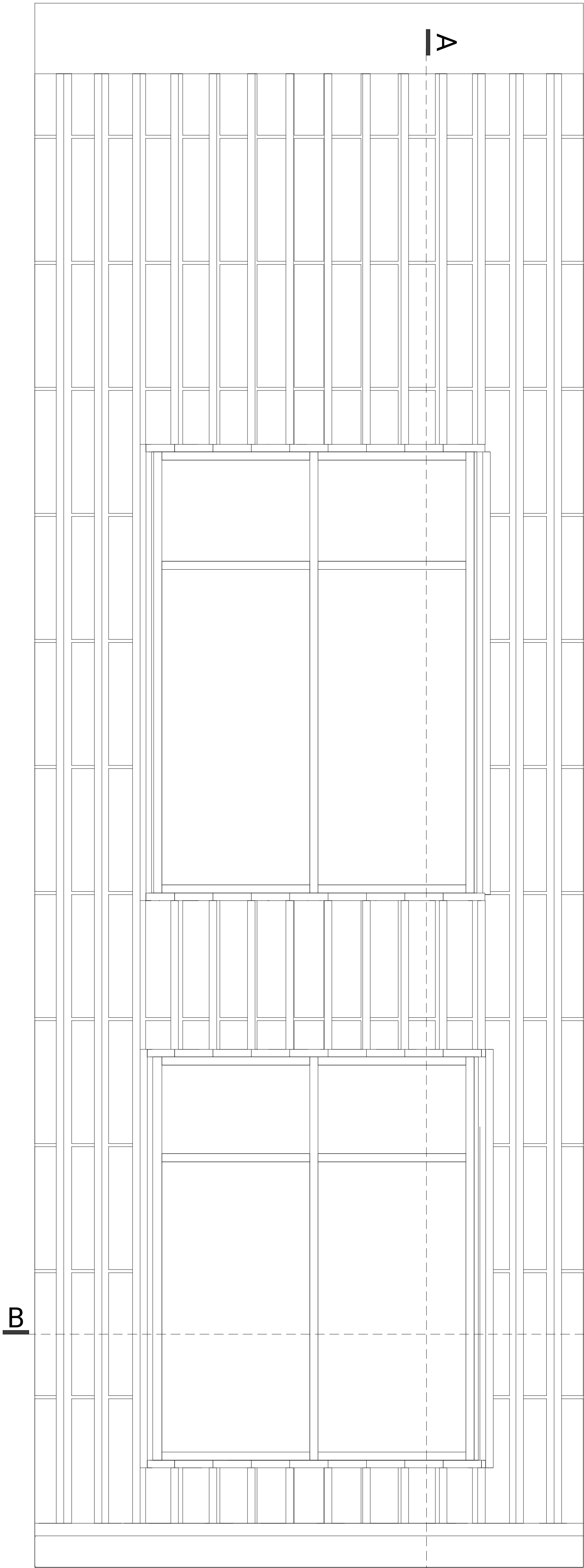
SECTION | 1 - 13



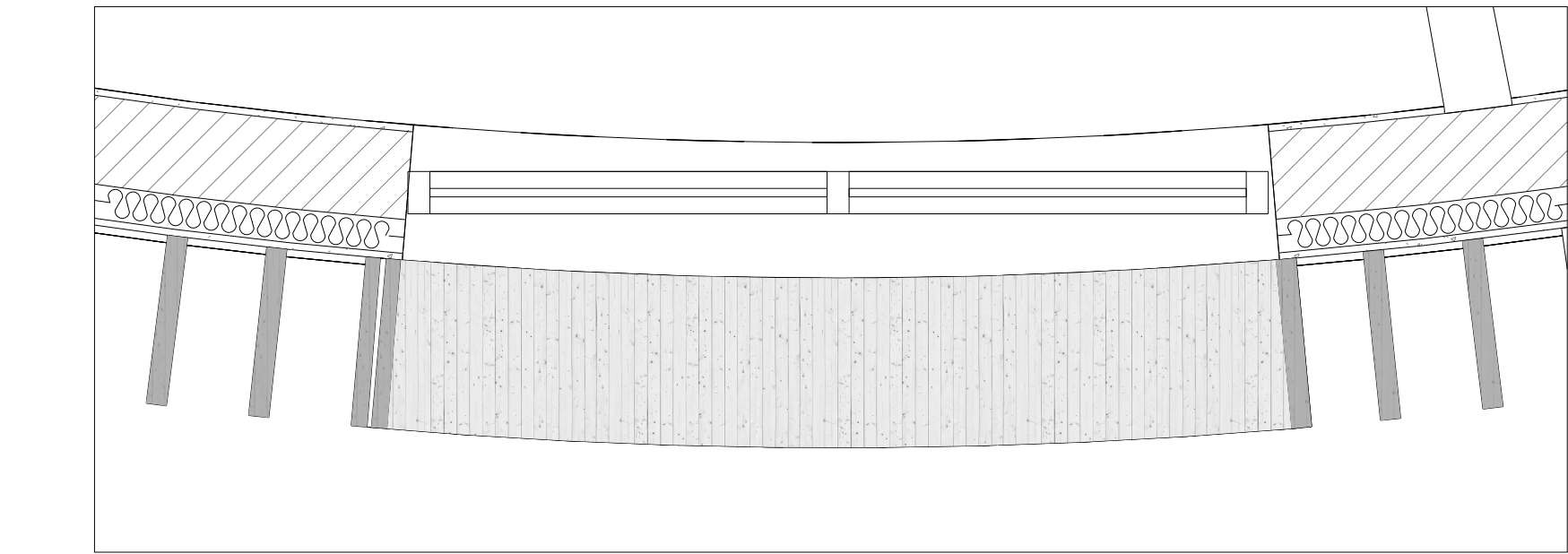
SECTION | A - H & D - H



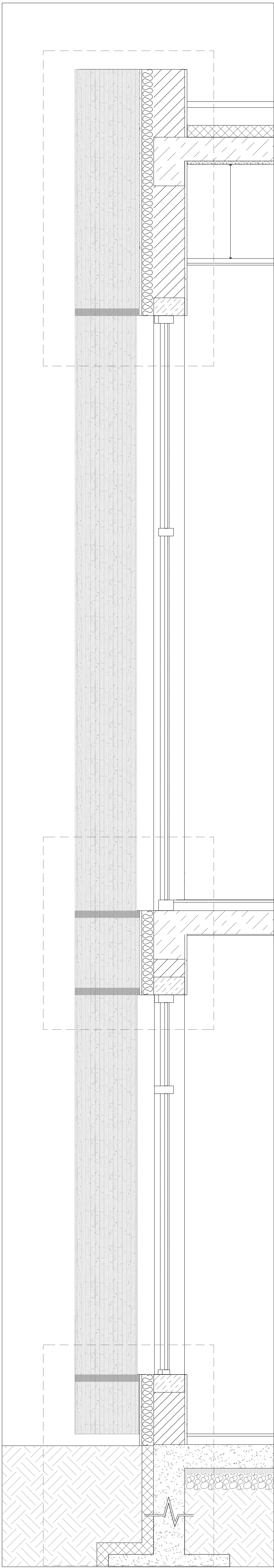
System Section



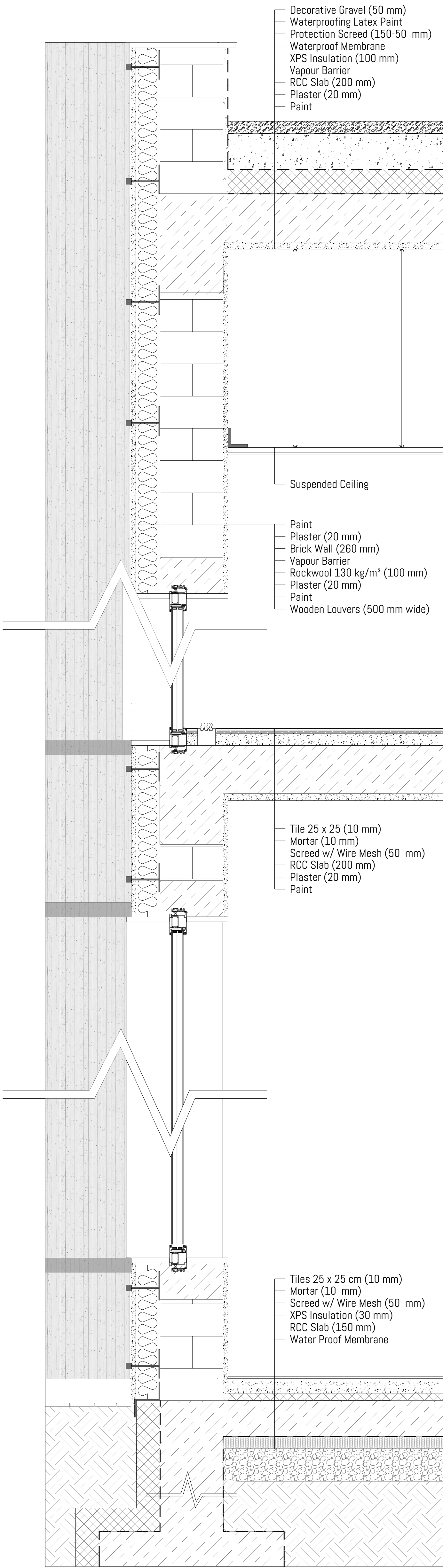
Facade 1:20



Section B-B | 1:20

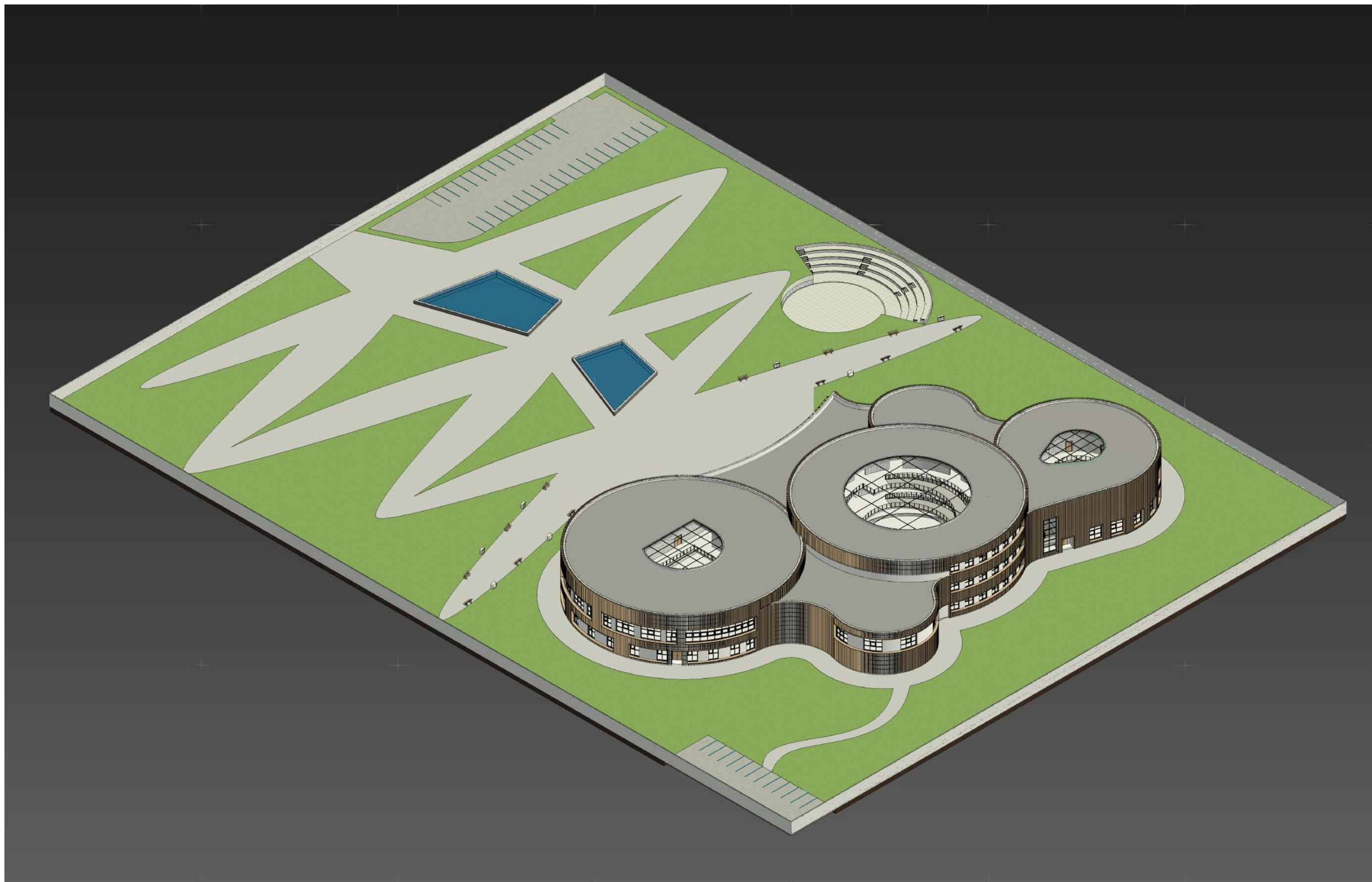


Section A-A | 1:20

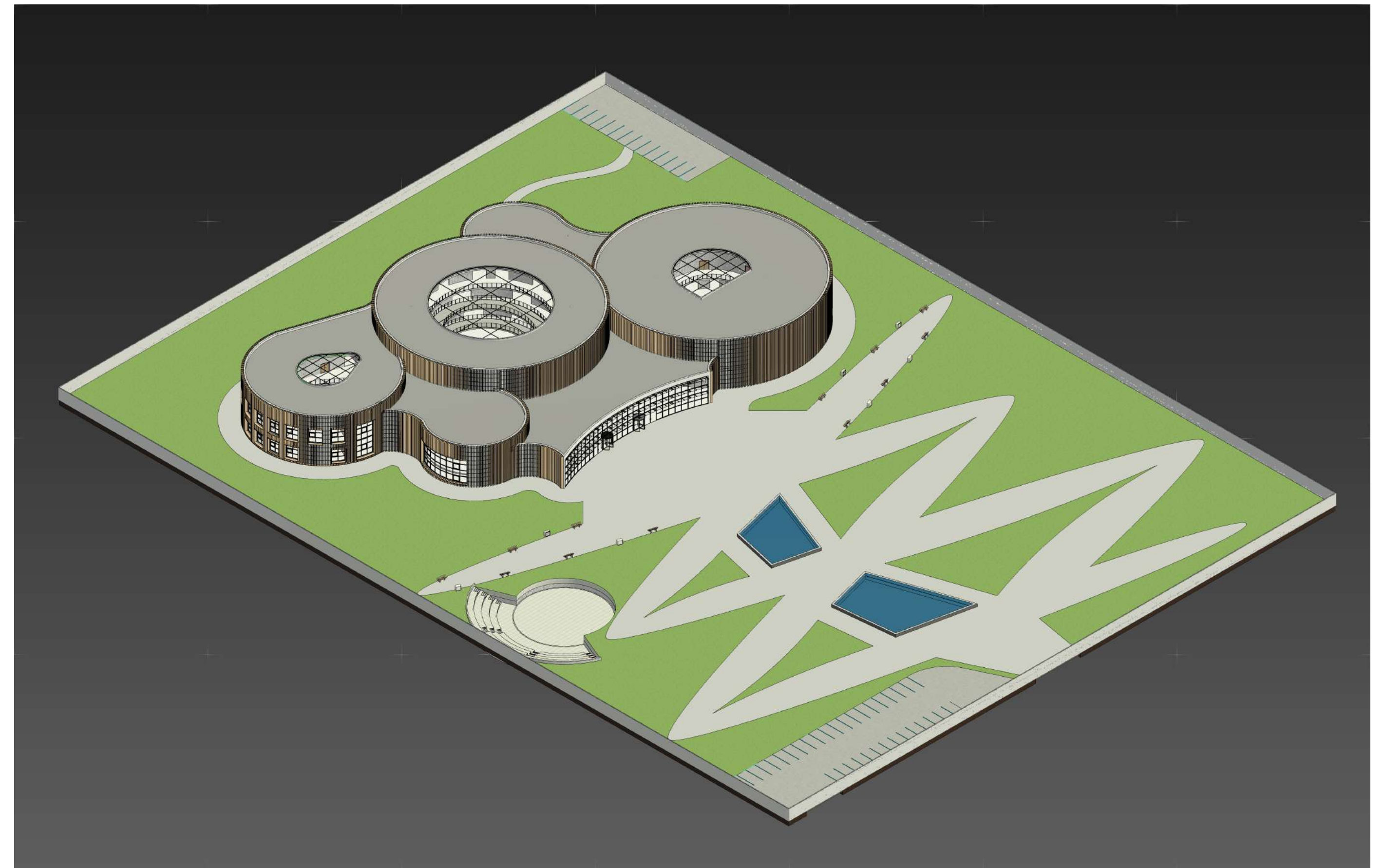


Callout Details | 1:10

AXONOMETRICS



North - East



South - West

RENDERS



RENDERS

